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RILEVAMENTO DELLE INTRUSIONI CON ALGORITMI DI OUT-OF-DISTRIBUTION: UNO STUDIO SPERIMENTAL

INTRUSION DETECTION WITH OUT-OF-DISTRIBUTION ALGORITHMS (AN EXPERIMENTAL STUDY)

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ABSTRACT

Intrusion detection systems are crucial for protecting modern computer networks from malicious attacks. Traditional intrusion detection methods often use supervised or unsupervised machine learning techniques applied directly to tabular network traffic data. While supervised classifiers, such as decision tree ensembles, frequently perform well when labelled attack data is available, they may suffer when anomalous or malicious samples are few or unavailable during training. Recent research has investigated the usage of Out-of-Distribution (OOD) detection techniques and deep learning models to address these issues.

This thesis focuses on the use of image-based OOD detection algorithms for anomaly detection in intrusion detection datasets. The study specifically investigates the transformation of tabular network traffic data into visual representations using the DeepInsight technique, which allows the usage of deep neural networks, which are initially developed for applications involving image processing. The study compares different deep learning-based anomaly detection models, including Autoencoder (AE), Variational Autoencoder (VAE), and Deep Support Vector Data Description (DeepSVDD), to DeepInsight-generated images.

Experiments are carried out on the SDN20 and BAIoT Mirai intrusion detection datasets. The models are tested in two experimental settings: Two-Class, which uses both normal and attack samples during training, and one-class, which trains models only on normal network traffic. For comparison, tabular methods, in particular Random Forest and Isolation Forest, are also evaluated.

The experimental results show that reconstruction-based deep learning models outperform DeepInsight-generated images, specifically the Autoencoder and Variational Autoencoder models. These models consistently perform better than the hypersphere-based DeepSVDD technique on both datasets. The findings further reveal that, while tabular supervised classifiers perform excellently when labelled attack data is available, deep learning-based anomaly detection models offer a flexible approach for detecting abnormal network traffic when only normal training data is available.

Overall, the findings indicate that implementing tabular-to-image transformation techniques with deep learning-based OOD detection algorithms offers a practical way forward for improving intrusion detection systems and identifying previously unseen network attacks.

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1. INTRODUCTION

1.1 BACKGROUND

Intrusion Detection Systems (IDS) play a very important role in protecting networks and other systems by identifying unauthorized access and abnormal behaviors. Identifying anomalies in network data has become increasingly important as attacks get more sophisticated. In Intrusion Detection Systems (IDS), anomaly detection mainly focuses on finding deviations from normal patterns, which may lead to unauthorized or malicious activities (Ahmed et al., 2016; Sommer & Paxson, 2010). Binary classifiers are frequently used in conventional intrusion detection techniques to differentiate between normal and abnormal behavior by using labels in datasets. Even if these techniques are successful, they rely significantly on the availability of anomalous data that has been properly categorized or labeled.

However, unsupervised or supervised approaches are frequently used since it is challenging to gather enough anomalous data for training in real-world scenarios (Liu et al., 2008; Breiman, L. 2001). The goal of these techniques is to model the system's normal behavior and identify any deviations as possible intrusions. Moreover, Conventional IDS methods, for example, rule-based systems or signature-based, often struggle while discovering novel or zero-day attacks (Ceccarelli & Zoppi, 2023), which are unobserved as of yet and therefore have no predefined signatures.

Out-of-distribution (OOD) detection has gained popularity as a solution for anomaly detection issues in recent years to overcome these limitations. Although OOD techniques are commonly used in domains like image processing and computer vision (Lee, K., Lee, K., Lee, H., & Shin, J. 2018; DeVries, T., & Taylor, G. W. 2018), their use with tabular data, such as intrusion detection datasets, is yet largely unexplored. To apply image-based OOD algorithms to tabular data, structured information has to be efficiently transformed into formats that these models can use. OOD algorithms are introduced to identify the data points that deviate from the expected distribution, aligning with the requirements of intrusion detection systems to uncover unexpected or novel intrusions.

Additionally, there is a trend in OOD detection to transform tabular data into visual representations in order to make use of deep learning models, particularly convolutional neural networks (CNNs), which have demonstrated exceptional performance in image-based classification tasks. This technique provides an innovative approach to use visual processing methods on non-image data, which might enhance Intrusion Detection Systems (IDS) ability to identify anomalies more efficiently (Zhu et al. 2021). However, the ongoing research continues to explore how these methods compare in effectiveness to traditional tabular-data models.

In particular, an important development in this field involves the utilization of DeepInsight, a framework created to convert tabular data into images. This enables the use of Convolutional Neural Networks (CNNs), typically employed in image recognition, to identify anomalies in tabular data like that utilized in intrusion detection. DeepInsight's transformation of tabular data into images facilitates the use of CNNs, known for their capacity to capture spatial and hierarchical patterns (Sharma et al. 2019).

This thesis explores the application of OOD detection algorithms, commonly used in image processing, to the domain of tabular data for intrusion detection. By observing the possible ways of using visual OOD detection algorithms and transforming tabular data into image-based data. The purpose of this study is to provide insight into their usefulness and ability in the real-world applications of cybersecurity.

1.2 PROBLEM STATEMENT

Identifying malicious activities in network traffic is a crucial component of modern cybersecurity approaches, and Intrusion Detection plays a vital role in this process. Supervised learning models rely on large, labeled datasets containing normal and anomalous behaviors. However, acquiring enough labeled anomalous data is difficult in real-world scenarios because intrusion events are rare and cyberattacks constantly change (Breiman, L. 2001). The lack of data hinders the ability of supervised learning models to perform well, reducing their effectiveness in detecting new or unknown intrusions.

Researchers have investigated unsupervised and supervised methods to address this limitation, aiming to identify anomalies by understanding patterns in normal data (Liu et al., 2008). Out-of-distribution (OOD) detection is one such approach; the main aim of this technique is to identify data points that deviate from the expected distribution in image processing (Hendrycks, D., & Gimpel, K., 2017). In the context of structured tabular data, commonly employed in intrusion detection systems, out-of-distribution (OOD) detection approaches have not received enough attention and remain unexplored and underdeveloped. In contrast, their usage has been generally effective in unstructured data domains such as computer vision.

Moreover, there has been a recent tendency to transform tabular data into images in order to utilize deep learning models like Convolutional Neural Networks (CNNs), which have demonstrated potential in specific situations. However, thorough research on its suitability for intrusion detection datasets has not been carried out yet. There is a gap in the current knowledge due to the absence of thorough research comparing conventional tabular-based models with image-based OOD detection methods for anomaly detection in this field.

The purpose of this study is to find out how effectively image-based OOD detection algorithms work with tabular data to detect intrusions. Specifically, the main goal of this thesis is to investigate whether converting tabular intrusion detection datasets into image-based and using OOD algorithms intended for visual processing can enhance the detection of anomalous activity in comparison to conventional tabular-based methods.

1.3 OBJECTIVES OF STUDY

The main goal of this research is to examine how anomaly detection in tabular intrusion detection datasets can be accomplished using Out-of-Distribution (OOD) detection algorithms, which are typically employed in image processing. In particular, the study seeks to accomplish the following goals:

1. To evaluate the possibility and efficiency of transforming tabular data intrusion detection into image data for use with image-based Out-of-distribution (OOD) detection techniques.

2. To assess the performance of different Out-of-distribution (OOD) detection techniques, including, but not limited to, Deep Support Vector Data Description (DeepSVDD), Autoencoders (AE), Variational Autoencoders (VAE), and other Generative Adversarial Networks (GAN)-based models, when applied to converted image data from intrusion detection datasets.

3. To evaluate these models' performance in two separate experimental environments. In the first environment, models are trained using both normal and attack samples for Two-Class; in the second environment, models are trained using just normal traffic data for One-Class.

4. To evaluate the efficiency of traditional tabular-based anomaly detection methods (like Isolation Forest and Random Forest) against Image-based Out-of-distribution (OOD) detection methods in the context of intrusion detection tasks.

5. To evaluate the advantages, drawbacks, and practical consequences of implementing image-based Out-of-distribution (OOD) algorithms in intrusion detection, particularly in terms of their scalability, robustness, and applicability in real-world scenarios.

To further insights into image-based out-of-distribution (OOD) detection methods, which can be an alternative to traditional tabular data methods, possibly enhancing anomaly detection techniques in cybersecurity

1.4 RESEARCH QUESTIONS

The main objective of this study is to examine how well Out-of-Distribution (OOD) detection algorithms function when applied to tabular datasets for intrusion detection. The following research questions have been developed to guide the study.

1. **How effective are tabular machine learning algorithms for detecting intrusions on tabular network traffic data?**

   When labelled data is available, tabular supervised machine learning models, particularly Random Forest, perform exceptionally well in intrusion detection. The Random Forest model performed nearly flawlessly across both datasets, demonstrating its exceptional ability to learn patterns from tabular network traffic data. The unsupervised Isolation Forest model, by contrast, performed significantly worse, underscoring the importance of labelled data for achieving high detection accuracy in traditional approaches.

2. **Can tabular network traffic data be effectively transformed into image representations for deep learning models?**

   Yes, the results show that tabular network traffic data can be effectively transformed into visual representations by using the DeepInsight approach. This change facilitates the usage of deep learning models designed for image processing applications. The high performance of deep learning models trained on these picture representations demonstrates that important spatial correlations between features may be captured, making this approach appropriate for intrusion detection applications.

3. **What is the comparison between the performance of OOD detection algorithms using image representations of tabular data and that of conventional anomaly detection algorithms applied directly to tabular data?**

   The research shows that out-of-distribution (OOD) detection algorithms utilized on image representations yield effective results, particularly in cases involving complex data patterns. Nevertheless, conventional tabular models such as Random Forest stand out for their efficiency across the datasets. Transforming tabular data into images allows for leveraging the strengths of deep learning models, which could enhance detection accuracy for high-dimensional, nonlinear data.

4. **Which algorithms for Out-of-Distribution (OOD) detection, including Deep Support Vector Data Description (DeepSVDD), Autoencoders (AE), and Variational Autoencoders (VAE), are the most efficient at identifying anomalies in datasets related to intrusion detection?**

The findings show that deep learning-based models, particularly reconstruction-based approaches like Autoencoder and Variational Autoencoder, perform similarly to classical supervised models in many scenarios. While Random Forest performs well in a supervised environment, deep learning models offer a more adaptable framework that can be used in both supervised and one class environments. This makes them better suited to real-world scenarios in which labelled data may not always be available.

5. **What are the differences in model performance between supervised (two-class) and one-class environments?**

   The experimental results reveal that models perform better in the supervised context than in the one-class scenario, because they can train on both normal and attack samples. However, in the one-class context, reconstruction-based models such as Autoencoder and Variational Autoencoder continue to perform well by learning the distribution of normal data and detecting deviations. This indicates their ability to detect anomalies even when attack data is not available for training.

6. **How do the selected algorithms perform across different intrusion detection datasets, such as SDN20 and BAIoT_mirai, in terms of accuracy, robustness, and scalability?**

   Our analysis revealed that the performance varied considerably among datasets due to variations in feature dimensions and class distributions. OOD algorithms showed greater resilience with SDN20 and BAIoT_mirai, especially when dealing with normal and abnormal data scenarios. Autoencoder and Variational Autoencoder excelled on datasets that exhibited distinctly defined normal patterns; both models demonstrated competitive capabilities in capturing intricate structures. Nevertheless, DeepSVDD showed inconsistencies across datasets, adapting more effectively to the organized setup of BAIoT_mirai than to SDN20. These results underscore the significance of matching algorithm selection with dataset features for effective anomaly detection.

7. **What are the most suitable metrics for assessing the effectiveness of OOD detection algorithms in the context of intrusion detection, and how do these metrics indicate practical relevance in real-world scenarios?**

   Metrics like Accuracy, Precision, Recall, and Overall F1-Score presented a thorough understanding of how the model performed, ensuring a balance between sensitivity and specificity.

   Additionally, we prioritized practical metrics such as computational efficiency and inference time to determine the feasibility of deploying the algorithms in the real-world. We evaluated computational efficiency by monitoring resource usage, including memory and processing power, during both model training and testing phases. Variational Autoencoders (VAEs) and Autoencoders (AE) showed relatively low resource requirements, making them well-suited for application in environments with limited resources. Although DeepSVDD was somewhat more demanding in terms of resources, it was fine-tuned for situations that necessitate greater flexibility.

8. **What are the real-world consequences of implementing image-based out-of-distribution (OOD) for intrusion detection systems in terms of computational expenses, complexity of deployment, and scalability?**

   Although algorithms for image-based out-of-distribution detection demonstrated great accuracy, they faced challenges related to computational demands and deployment difficulties. While it was achievable to scale up to extensive systems using optimized architectures and cloud deployments, it is crucial to consider resource limitations for them to be practically implemented.

1.5 SCOPE OF THE STUDY

The scope of the study focuses on examining the suitability and efficacy of Out-of-Distribution (OOD) detection algorithms for identifying anomalies in intrusion detection systems (IDS). For this analysis, two datasets are employed: "SDN20" (Meidan, 2018) and "BAIoT_mirai" (Alharbi, 2020), which illustrate conventional network intrusions and contemporary Internet of Things (IoT) attacks. These datasets offer a range of scenarios, from structured, clearly defined attack patterns to intricate, real-world.

The research assesses Random Forest as a supervised algorithm that utilizes labeled data to distinguish between normal and anomalous network traffic. Serving as a benchmark, Random Forest shows how traditional classification methods perform in terms of intrusion detection. At the same time, Isolation Forest is examined as an unsupervised algorithm for anomaly detection. Its ability to identify anomalies without the need for labeled data is especially pertinent in practical situations where acquiring labeled anomalous data can be difficult.

The research evaluates the effectiveness of three algorithms for anomaly detection: Deep Support Vector Data Description (DeepSVDD) (Ruff, et al. 2018), Autoencoders (AE) (Hawkins, D. M., et al. 2002), and Variational Autoencoders (VAE) (Kingma, D. P., & Welling, M. 2013). These algorithms are evaluated on tabular data, which is transformed into image formats using DeepInsight. This dual methodology allows for the investigation of the potential advantages of using image-based out-of-distribution (OOD) detection for structured tabular datasets, an area that is still relatively unexplored in present-day research.

The study underlines the importance of evaluating these algorithms from multiple perspectives, such as their accuracy, resilience, and computational efficiency. Specifically, inference time is identified as an essential measure for evaluating the practicality of implementing these algorithms in real-world cybersecurity backgrounds. Furthermore, the research investigates how resilient the algorithms are to challenges presented by datasets, including feature dimensionality and class imbalance.

While supervised classification techniques such as Random Forest or Isolation Forest provide a solid reference point for evaluation, the research primarily emphasizes out-of-distribution

(OOD) detection methods, particularly in situations where data is transformed image-based. This comprehensive strategy guarantees that the study covers a broad spectrum of practical applications, spanning from cases with thoroughly labeled datasets to those that depend on unsupervised detection methods, and then image-based.

The study mainly concentrates on unsupervised and supervised techniques, highlighting their ability to surpass conventional methods in contexts with unusual or unlabelled anomalous data as a significant aspect. To emphasize the exploration of the advantages and drawbacks of OOD detection methods within the chosen domain, supervised classification techniques are deliberately left out.

Additionally, the performance of the suggested intrusion detection models under two different experimental environments is evaluated in this thesis. The models are trained using both normal and attack samples from the training dataset in the first scenario, which is referred to as the Two-Class scenario. In the second scenario, defined as the One-Class scenario, the models are tested on their ability to identify malicious or anomalous behaviour after being trained only with normal samples. This comparison enables the study to explore the differences in model performance between training with labelled data and training with only normal data.

Above all, the research is confined to fixed datasets and does not encompass live traffic assessments. Nevertheless, the knowledge acquired in this study aims to aid the implementation of these algorithms in operational settings. This involves addressing limitations in computational resources, scalability, and evolving network threats, thereby ensuring that the results are pertinent to contemporary cybersecurity frameworks.

1.6 STRUCTURE OF THE THESIS

The report is structured to provide a comprehensive and systematic exploration of the research topic, beginning with foundational concepts and leading to detailed evaluations and conclusions. The structure is outlined as follows:

Chapter 1: Introduction

This chapter sets the stage for the research by presenting the background and motivation for the study. It outlines the problem statement, research objectives, research questions, scope of the study, contributions, and significance. This chapter establishes the context for the research and highlights its importance in addressing contemporary challenges in intrusion detection systems.

Chapter 2: Literature Review

The second chapter provides a detailed review of existing literature related to anomaly detection and intrusion detection systems. It covers supervised, unsupervised, and Out-of-Distribution (OOD) detection algorithms and their applications. The chapter also discusses data representation techniques, such as the conversion of tabular data to image formats, and highlights research gaps that the study aims to address.

Chapter 3: Methodology

This chapter explains the research design and methods used in the study. It describes the datasets selected for evaluation (SDN20, BAIoT_mirai, and NSL-KDD) and the preprocessing steps applied. The methodology for applying supervised (Random Forest), unsupervised (Isolation Forest), and OOD detection algorithms (DeepSVDD, AE, and VAE) is detailed. Additionally, the chapter outlines the evaluation metrics used to assess performance and the approach to comparative analysis.

Chapter 4: Results and Analysis

The fourth chapter presents the findings of the study, including the performance of each algorithm across the selected datasets. It provides detailed analyses of the results, highlighting patterns, strengths, and limitations of the algorithms. This chapter also explores the impact of data representation (tabular vs. image-based) on algorithm performance and discusses practical implications for real-world deployment.

Chapter 5: Discussion/Main Findings

This chapter interprets the results in the context of the research objectives and questions. It discusses the broader implications of the findings, particularly for the design and deployment of intrusion detection systems. Challenges and limitations encountered during the research are also addressed, along with potential solutions and improvements.

Chapter 6: Conclusion and Future Work

The final chapter summarizes the key findings of the study and their contributions to the field of anomaly detection and cybersecurity. It emphasizes the significance of the research and provides recommendations for future work, including directions for improving algorithms, datasets, and methodologies.

1.7 CONTRIBUTION AND SIGNIFICANCE

This research makes several significant contributions to the field of anomaly detection and intrusion detection systems (IDS), particularly by exploring the application of supervised, unsupervised, and Out-of-Distribution (OOD) detection algorithms in different scenarios. The findings offer both theoretical insights and practical implications. Practically, it provides actionable insights for deploying efficient and scalable intrusion detection systems, addressing real-world challenges in network security and IoT environments. The findings not only improve our understanding of existing methods but also inspire future innovations in anomaly detection and IDS design.

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2. LITERATURE REVIEW

Modern cybersecurity frameworks rely heavily on Intrusion Detection Systems (IDS) to identify and prevent unauthorized access to networks and systems. Researchers have thoroughly studied a wide range of anomaly detection strategies, from advanced Out-of-Distribution (OOD) detection algorithms to more traditional supervised and unsupervised approaches. This review summarises the prior literature, emphasising its applications, limitations, and related work.

2.1 TRADITIONAL MODELS FOR TABULAR INTRUSION DETECTION:

In the context of cybersecurity, Traditional approaches for tabular intrusion detection are crucial for evaluating structured datasets derived from network activities. Because of their interpretability and efficiency in processing feature-driven representations of intrusion patterns. XGBoost and Random Forest are two popular algorithms for structured tabular data. These models exhibit strong classification performance when trained on labeled datasets comprising normal and anomalous instances. Random Forest was presented by (Breiman, L. 2001) as an extremely accurate and interpretable ensemble model that works well with high-dimensional datasets. Nevertheless, supervised techniques rely significantly on labelled data, which is sometimes lacking in real-world intrusion detection scenarios.

Unsupervised techniques like Isolation Forest (IsoForest) have become more and more popular in response to the lack of labeled data. Isolation Forest (IsoForest) is suitable for scenarios when prior information on anomalies is unavailable, since it isolates anomalies by randomly splitting the data. Despite its effectiveness, Isolation Forest (IsoForest) struggles with high-dimensional and complex datasets where anomalies could be difficult to separate from the normal data (Liu et al., 2008).

2.2 OUT-OF-DISTRIBUTION DETECTION IN MACHINE LEARNING:

OOD detection finds data points that significantly deviate from the training distribution, which is an extension of traditional anomaly detection. OOD detection was first developed for image recognition and natural language processing (Krizhevsky et al., 2012), but now OOD detection has made its way into tabular data through innovative approaches.

A study by Lee et al. (2018) indicates that most OOD detection research has been conducted in computer vision, where models are trained on image datasets and tested on adversarial or unknown data. These methods frequently use probabilistic modelling and deep neural networks to differentiate between samples that are in-distribution and those that are not.

Because network traffic patterns can change over time and new attack types may appear, OOD detection is especially helpful in the context of intrusion detection. OOD approaches can detect previously unseen attacks by identifying abnormal traffic as out-of-distribution samples by modelling typical traffic as the in-distribution.

2.3 TRANSFORMATION OF TABULAR DATA INTO IMAGE REPRESENTATIONS:

(Sharma et al. 2019) introduced DeepInsight, a transformational method that allows convolutional neural networks (CNNs) to process structured datasets by transforming tabular data into 2D image representations. DeepInsight maintains feature relationships by employing techniques such as t-SNE to map data into a 2D space. This enables CNNs to use spatial dependencies to detect anomalies. This method significantly improves the detection of complex patterns by bridging the gap between tabular and image-based approaches.

CNNs trained on image representations of tabular data perform better than conventional machine learning models, as shown by (Zhu et al. 2021). The transformation highlights the potential of image-based approaches for intrusion detection by making it possible to detect spatial and hierarchical patterns that are hard to identify in raw tabular formats.

2.4 LATENT REPRESENTATION AND HYPERSPHERE-BASED OOD MODELS:

OOD detection focuses on identifying instances that deviate from the normal data distribution. These techniques employ deep learning to manage the complexity of modern datasets.

Normal data is mapped into a compact hypersphere in the latent space by DeepSVDD, which was first presented by (Ruff et al., 2018). It works very well with structured datasets that have well-defined normal patterns. However, anomalies are under-represented in imbalanced datasets, which presents challenges.

Autoencoders use compressed latent representations to reconstruct input data. Reconstruction errors can be used to detect anomalies, according to research by (Hawkins et al. 2002). Although useful, autoencoders need careful threshold setting and are susceptible to noise. Though they are useful, autoencoders need careful threshold tuning because they are noise-sensitive.

Later on, (Zong et al. 2018) introduced the Deep Autoencoding Gaussian Mixture Model (DAGMM), which combines autoencoding and probabilistic modelling to increase anomaly detection performance.

(Meidan et al. 2018) used deep autoencoders to detect IoT botnets and reported strong results on the N-BaIoT dataset, illustrating the practical importance of deep anomaly detection for network security.

(Doersch, 2016) presented a full instruction on VAEs, emphasising their generative capabilities. VAEs are especially useful for modelling complex data distributions. However, VAEs may produce blurrier reconstructions and sometimes weaker anomaly separation compared to deterministic autoencoders.

2.5 APPLICATION OF OOD DETECTION IN INTRUSION DETECTION

OOD detection methods have been effectively used to identify new and unseen attacks, increasing the adaptability of IDS.

Zero-Day Attack Detection:

OOD detection plays a crucial role in detecting zero-day attacks, as shown by Ceccarelli and Zoppi (2023), where no previous examples exist. IDS may effectively detect anomalies without depending on predefined attack signatures by modelling typical behaviour and utilizing OOD techniques. This provides resilience against evolving threats.

Deep Learning Techniques:

Traditionally, CNNs have been employed for image-based OOD detection tasks. This application was expanded to tabular datasets by (Zhu et al. 2021), demonstrating that CNNs can identify meaningful patterns by converting tabular data into image representations.

Libraries such as PyOD and Alibi Detect offer adaptable frameworks for applying OOD detection methods to both structured and unstructured data, making it easier to integrate them into IDS.

2.6 CURRENT RESEARCH FINDINGS

Recent developments in OOD detection demonstrate how successful it is at detecting intrusions:

DeepInsight Framework: (Sharma et al. 2019) showed that by preserving feature relationships in a 2D space, DeepInsight dramatically enhances model performance on tabular datasets. This enables CNNs to identify complex patterns, increasing anomaly detection accuracy.

CNN-Based Anomaly Detection: CNNs are superior to tabular machine learning models, as demonstrated by (Zhu et al. 2021), who validated their efficiency in identifying anomalies from image representations of tabular data.

OOD Data Generation: As noted by (Ceccarelli and Zoppi, 2023), generating OOD data is crucial for IDS, especially when it comes to identifying new attacks like zero-day intrusions. By using this technique, IDS can adapt more easily to evolving attack environments.

2.7 SUMMARY

The research currently in publication highlights how intrusion detection systems (IDS) have developed from traditional supervised and unsupervised approaches to more sophisticated Out-of-Distribution (OOD) detection techniques. According to (Breiman, 2001), supervised learning techniques like Random Forest and XGBoost have proven to perform well in structured environments with labeled data, offering reliable and interpretable classification.

Supervised learning approaches, such as Random Forest and XGBoost, have demonstrated strong performance in structured environments with labeled data, providing robust and interpretable classification (Breiman, 2001). However, their applicability in dynamic, real-world scenarios where anomalous data may be scarce or undefined is limited by their reliance on labelled datasets.

In contrast, unsupervised techniques such as Isolation Forest are appropriate for environments with novel or evolving attack patterns because they can identify anomalies without the need for labelled examples (Liu et al., 2008). Despite their adaptability, these techniques frequently struggle to handle overlapping or high-dimensional datasets, which can reduce their efficacy. Due to the limitations of both supervised and unsupervised approaches, OOD detection techniques were developed. These techniques concentrate on identifying data points that deviate from the normal training distribution (Ruff et al., 2018).

The development of approaches such as DeepInsight has bridged the gap between tabular and image-based data analysis, allowing the implementation of powerful convolutional neural networks (CNNs) on tabular-based datasets. DeepInsight preserves feature relationships by converting tabular features into 2D grids, enabling CNNs to identify complex patterns and outliers that are difficult to spot in raw tabular-based data (Sharma et al., 2019). This development, together with other techniques such as t-SNE and dimensionality reduction (van der Maaten & Hinton, 2008), has greatly improved OOD algorithms' capacity to handle structured data.

OOD approaches in intrusion detection have demonstrated significant potential to address challenges like zero-day attack detection, concept drift, and class imbalance. Research by (Ceccarelli and Zoppi 2023) has highlighted the adaptability of OOD approaches in detecting novel attacks without relying on pre-existing signatures, making them especially useful in dynamic and emerging network environments. Similarly, studies by (Sharma et al. 2019) and (Zhu et al. 2021) show how image-based representations effectively improve anomaly detection accuracy and generalization.

Despite significant progress in intrusion detection and OOD detection research, several challenges remain. One major challenge is the limited availability of labeled attack data, which restricts the applicability of supervised learning approaches. Additionally, network traffic data is often highly imbalanced, with normal traffic significantly outnumbering malicious samples, making it difficult for models to learn effective decision boundaries (He & Garcia, 2009).

The study of unsupervised and one-class methods is motivated by some limitations, for ex: Computational Efficiency. Although image-based methods are efficient, they need a lot of resources for training and data transformation, as explained by (Zarpelão et al. 2017) and (Sharma et al. 2019).

Moreover, recent developments in hybrid techniques, explainable AI, and lightweight IDS have broadened the scope of research in intrusion detection. To improve detection rates and lower false positives, hybrid models integrate the strengths of supervised and unsupervised approaches (Ashfaq et al., 2017). IDS transparency and trust are increased by explainable AI frameworks like SHAP (Lundberg & Lee, 2017), which offer insights into model decisions. Lightweight IDS systems are ideal for resource-constrained environments such as IoT networks, where computational efficiency is crucial (Zarpelão et al., 2017).

Also, the significance of addressing scalability and adaptability in IDS is emphasized in the literature. Scalable algorithms capable of real-time detection are crucial as network traffic volumes and complexity increase (Ahmed et al., 2016). Similarly, the ability to adapt to evolving attack patterns by drift detection techniques guarantees that IDS remains successful over time (Gama et al., 2014).

In summary, while traditional supervised and unsupervised methods remain fundamental to anomaly detection, OOD techniques and hybrid approaches provide substantial advances in dealing with modern intrusion detection challenges. This study intends to address significant scalability, adaptability, and performance gaps by integrating approaches such as DeepInsight and leveraging CNN capabilities. These findings establish a robust framework for comparing the effectiveness of traditional and modern methods in intrusion detection systems.

3

3. METHODOLOGY

This study utilizes a multi-step methodology to evaluate the efficiency of Out-of-Distribution (OOD) detection techniques on tabular intrusion detection datasets, generally used for image data. The methodology encompasses the selection of datasets, data processing, applying algorithms, and results assessment, as outlined below.

3.1 RESEARCH DESIGN

This research utilizes a quantitative experimental framework to investigate the efficiency of supervised, unsupervised, and Out-of-Distribution (OOD) detection algorithms in detecting anomalies within intrusion detection systems (IDS). The study adopts a methodical approach to tackle the primary goals and respond to the outlined research questions.

The framework integrates the assessment of various anomaly detection algorithms, encompassing supervised approaches like Random Forest, unsupervised strategies such as Isolation Forest, and out-of-distribution detection methods including Deep Support Vector Data Description (DeepSVDD), Autoencoders (AE), and Variational Autoencoders (VAE). This research investigates their relevance and efficiency across two different forms of data representation: Conventional tabular methods and Image-based representations produced through DeepInsight.

The study utilizes two datasets that represent a variety of intrusion detection conditions: SDN20 and BAIoT_mirai. These datasets were selected due to their significance in reflecting both conventional and contemporary cybersecurity issues, providing an extensive platform to assess and analyze the robustness, accuracy, precision, and efficiency of the chosen algorithms.

The research evaluates the algorithms using crucial performance indicators such as accuracy, Precision, Recall, F2-score, computational efficiency, and inference time. A comparative review is conducted to identify the advantages and disadvantages of each algorithm across various datasets and data formats. Special attention is given to practical applicability by evaluating the necessary computational resources and scalability.

The research methodology involves data preprocessing to maintain consistency and quality, then executing each algorithm with uniform configurations for reasonable comparisons.

Moreover, through methodical testing and analysis of results, the study offers practical insights into the implementation of anomaly detection algorithms within operational intrusion detection systems.

3.2 DATASET SELECTION

The datasets chosen for this research encompass a variety of real-world situations in intrusion detection systems (IDS). Each dataset captures unique features, enabling a thorough assessment of the algorithms in different situations. The selected datasets are SDN20 and BAIoT_mirai, highlighted for their significance in both conventional and modern cybersecurity environments.

1. **SDN20 Dataset:** The SDN20 dataset is specifically designed for Software-Defined Networking (SDN) contexts, documenting traffic behaviors across a variety of attack scenarios. This dataset is exceptionally useful for assessing algorithms in contemporary, adaptive networking environments where attacks have become progressively advanced. SDN20 includes traffic data representing both normal and malicious, with attributes that depict different network features, such as flow duration, packet sizes, and types of protocols. The dataset comprises 205,167 records and 78 features. It contains a combination of numerical and categorical information, including Protocol (a categorical variable) and diverse traffic flow metrics expressed as integers and floating-point values. The dataset's considerable dimensionality and variability offer a challenging but valuable environment for testing anomaly detection techniques. The preprocessing procedures for SDN20 involve:

   Handling Missing Values: Rows that contain missing or incomplete information are discarded to maintain the dataset's integrity.
   Label Encoding: Categorical features, like protocol types, are converted into numerical forms to ensure compatibility with machine learning models.
   Normalization: Feature values are adjusted to a standard range to enhance the performance of machine learning algorithms.

1. **BAIoT_mirai Dataset:** The BAIoT_mirai dataset focuses on IoT traffic, with a specific emphasis on attacks conducted by the Mirai botnet. This dataset illustrates actual IoT environments where devices with limited resources are especially vulnerable to botnet intrusions. It encompasses comprehensive network flow data, capturing both normal and abnormal traffic. The dataset comprises 75,164 records and 116 features. The majority of the features are numerical (float64), including metrics that describe the packet flow weights, means, and variances, while one column labeled multilabel acts as the target class. This dataset offers extensive insight into IoT traffic patterns, rendering it useful for evaluating anomaly detection systems. The main preprocessing steps involve:

   Handling Missing Values: Rows that contain missing or incomplete information are discarded to maintain the dataset's integrity.

Label Encoding: Categorical features, like protocol types, are converted into numerical forms to ensure compatibility with machine learning models.
Normalization: Feature values are adjusted to a standard range to enhance the performance of machine learning algorithms.

Each dataset will be split into training and testing subsets, making sure that normal and malicious data points are separated, which supports both supervised and unsupervised testing approaches. The datasets were chosen to ensure that the study addresses multiple real-world challenges in intrusion detection. By integrating datasets that include different attack types, complexities of features, and variations in data distributions, the study intends to thoroughly assess the robustness, accuracy, and scalability of the chosen algorithms. This varied selection ensures that the results of the research can be applied to a broad spectrum of cybersecurity contexts.

3.2.1 DATA PREPROCESSING:

The efficiency of intrusion detection depends on proper preprocessing. The following procedures were used in every experiment.

3.2.1.1 MISSING VALUE HANDLING:

Zeros were used to replace any missing values in the datasets. In addition to preventing training errors in downstream models, this step provides numerical stability.
In formal terms, for every feature vector x:

$$x_i = \begin{cases} 0 & \text{if missing} \\ x_i & \text{otherwise} \end{cases}$$

3.2.1.2 LABEL ENCODING:

Binary representations of the first multi-class labels were created:

0 = normal traffic
1 = anomaly or attacked

Both are supported by this conversion:

Supervised Anomaly Detection
One Class Intrusion Detection

3.2.1.3 FEATURE NORMALIZATION:

Z-score normalisation was used to standardise all tabular features:

$$x' = \frac{x - \mu}{\sigma}$$

where σ is the standard deviation, and μ is the feature mean.
Standardisation guarantees quicker convergence of neural networks and enhances distance-based modelling with balanced feature scaling.

3.2.1.4 PARADIGMS OF EXPERIMENTS:

ODD Two-Class Setting:

Both attack and normal data were used to train the models in the Two-Class environment, and were particularly used for Random Forest, Isolation Forest for evaluation purposes, and AE/VAE/DEEPSVDD Models only if needed. This configuration exemplifies a conventional IDS implementation.

ODD One-Class Setting:

Models were trained solely on normal traffic in the One-Class scenario and were particularly used for DeepSVDD, Autoencoder, and variational autoencoder. This configuration mimics real-world scenarios in which attack samples might not be accessible.

3.3 DATA CONVERSION: TABULAR TO IMAGE

A fundamental aspect of this research involves transforming tabular data into image formats (32x32 grayscale images) to utilize image-based anomaly detection techniques. This process was executed through the DeepInsight library, which maps tabular data features onto a two-dimensional grid, generating significant spatial connections among the features. This conversion assigned each feature to particular pixel positions by feature importance and correlations, making sure that the spatial depiction maintained the significant relationships present in the data. This conversion enables structured tabular data to be examined using convolutional neural networks (CNNs) and various other image-based models.

3.3.1 DEEPINSIGHT FRAMEWORK

To implement image-based out-of-distribution (OOD) detection techniques, we will convert tabular data into image formats. The DeepInsight library has been employed to turn individual data points into images by mapping features to specific pixel positions on a two-dimensional grid. This conversion process comprises:

Feature Mapping: Features are prioritized according to their correlations or significance. Each feature is allocated a specific spot on a two-dimensional grid, allowing closely related features to be positioned closer to one another to preserve meaningful spatial connections.

Normalization: Feature values are adjusted to a consistent range of [0, 1] to maintain uniformity among all features. This normalization helps achieve consistent intensity levels in the generated images, enabling improved learning by image-based models.

Image Creation: Every data point (row in the table) is visualized as an image, with pixel intensity levels reflecting the normalized values of the features. Grayscale images are produced for easier understanding, while RGB images are generated by mapping several features or dimensions to different color channels (Red, Green, Blue).

Features were assigned to a predetermined grid size; different grids were used for different experiments to evaluate the performance:

$$8 \times 8 — 16 \times 16 — 24 \times 24 — 32 \times 32$$

Data Augmentation: To enhance the diversity and reliability of the training dataset, augmentation methods like flipping, rotating, and scaling are implemented on the produced images. This guarantees that the models perform better with unseen data.

Every sample was transformed into an image with normalised feature values corresponding to pixel intensities. Convolutional architectures are made possible by this transformation.

3.3.2 JUSTIFICATION FOR CONVERSION

The motivation to transform tabular data into images originates from image-focused models, especially CNNs, in recognizing spatial patterns. In contrast to conventional anomaly detection techniques that analyze feature values on their own, image-based data representations enable models to identify complex interactions and relationships among features, which could enhance the effectiveness of anomaly detection.

3.3.3 CHALLENGES AND CONSIDERATIONS

Transforming tabular data into images involves various challenges:

Grid Design: It's essential to organize features on the grid in a manner that maintains the original relationships within the data.

Dimensionality: Datasets with high dimensions can result in large image sizes, which in turn can increase processing costs. Techniques for reducing dimensionality, such as Principal Component Analysis (PCA), are examined to tackle this problem.

Information Loss: Multiple efforts are made to reduce information loss during the process of mapping and transforming features, ensuring that essential characteristics of the data are retained.

3.3.4 BENEFITS OF IMAGE-BASED REPRESENTATIONS

Enhanced Feature Interactions: The spatial relationships among features provide extra contextual information, enhancing the interpretability of the model.

Compatibility with Advanced Models: Utilizing image representations facilitates the application of deep learning architectures like CNNs, which have shown exceptional results in out-of-distribution (OOD) detection challenges.

Uniform Data Representation: Transforming varied tabular datasets into images creates consistency in input formats, facilitating a cohesive pipeline.

This technique of transforming tabular data into images enables a unique exploration into the performance of image-based anomaly detection methods, connecting structured tabular datasets with deep learning techniques typically applied in computer vision tasks.

3.4 INTRUSION DETECTION ALGORITHMS ON TABULAR DATA

Traditional anomaly detection methods that rely on tabular data will be employed on the initial datasets as a reference point for comparison. The selected algorithms include:

3.4.1 RANDOM FOREST

This is a supervised ensemble learning approach that builds numerous decision trees and determines the classification by majority vote. Its capacity to manage high-dimensional datasets and its robustness against overfitting render it effective for intrusion detection applications. The Random Forest algorithm has been trained using labeled datasets (SDN20, BAIoT_mirai), which include both normal and attack data, to evaluate its classification efficiency.

3.4.2 ISOLATION FOREST

Isolation Forest is an unsupervised anomaly detection technique that uses random partitioning to isolate data points to find anomalies. In this approach, outliers are considered as points that are easier to isolate (need fewer partitions). The approach works well for finding unseen attacks in high-dimensional datasets since it relies on feature randomization and has a low computing cost. The Isolation Forest algorithm has been trained as an unsupervised algorithm using datasets (SDN20, BAIoT_mirai).

3.5 OOD ALGORITHMS FOR VISUAL PROCESSING

After the transformation of data, the research will apply image-based out-of-distribution (OOD) detection techniques to the generated images. The chosen OOD methods include:

3.5.1 DEEP SUPPORT VECTOR DATA DESCRIPTION (DEEPSVDD)

This method finds a hypersphere that encompasses normal data in the feature space, while outliers are situated beyond this hypersphere. It proves especially useful in scenarios where the distributions of normal and abnormal data intersect.

In feature space, DeepSVDD learns a hypersphere that encloses typical data.

The goal is:

$$\min_{W} \frac{1}{n} \sum_{i} \| f_W(x_i) - c \|^2$$

where:

f_W is the neural network

c is the centre hypersphe

Anomalous samples are those that are far from the center.

3.5.2 AUTOENCODERS (AE)

Autoencoders are neural networks designed to reconstruct input data. Anomalies are identified through the reconstruction error, with greater errors signifying outliers.

Architecture:

The encoder is Conv = Pool = Dense,

The latent space,

The decoder is: Dense → Deconv,

Scores of anomalies,

$$\text{Error}(x) = \| x - \hat{x} \|^2$$

Anomalies are indicated by higher reconstruction error.

Advantages: Detects/captures complicated patterns in data and is well-adapted.

Limitations: sensitive to noise and needs thresholds to be carefully adjusted.

3.5.3 VARIATIONAL AUTOENCODERS (VAE)

The variational Autoencoders (VAE) enhance AEs by using probabilistic latent spaces. They learn to create data that is compatible with the input distribution, with anomalies identified based on their deviation from the learned distribution.

By acquiring a probabilistic latent distribution, the VAE expands on AE.
This is the loss function:

$$\mathcal{L} = \text{Reconstruction Loss} + \text{KL Divergence}$$

Although VAEs offer smoother latent representations, they might result in less distinct anomalies.

Advantages: Efficient in capturing complicated distributions of data.

Limitations: Computationally demanding and may not be able to handle large-scale data.

3.6 EVALUATION METRICS

A variety of measures were employed to provide a thorough evaluation.

Accuracy:

Accuracy measures the overall proportion of correctly classified instances among all evaluated samples. In the context of intrusion detection, accuracy represents how effectively the model can correctly identify both normal network traffic and malicious traffic.

Accuracy is calculated as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

where TP represents true positives, TN represents true negatives, FP represents false positives, and FN represents false negatives.

A higher accuracy value indicates that the model performs well in correctly classifying network traffic samples.

Precision:

Precision measures the proportion of correctly identified attack samples among all samples predicted as attacks by the model. It indicates how reliable the intrusion detection system is when it flags network traffic as malicious. A higher precision value means that the model generates fewer false alarms.

Precision is calculated as:

$$\text{Precision} = \frac{TP}{TP + FP}$$

where TP represents true positives, and FP represents false positives.

Recall:

Recall, also known as the detection rate or sensitivity, measures the proportion of actual attack samples that are correctly identified by the model. In intrusion detection systems, recall is important because it reflects the ability of the system to detect malicious activities and avoid missed attacks.

Recall is calculated as:

$$\text{Recall} = \frac{TP}{TP + FN}$$

where TP represents true positives, and FN represents false negatives.

F1-Score:

The F1-score is the harmonic mean of precision and recall. It provides a balanced evaluation of the model's performance by considering both the detection capability and the rate of false alarms. This metric is particularly useful when evaluating intrusion detection models where both false positives and false negatives are important.

F1-Score is calculated as:

$$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$$

3.7 EXPERIMENT SETUP

Out-of-distribution (OOD) detection approaches in Intrusion Detection using image-based data obtained from tabular data, which was the goal of the experimental setup design for this work. This configuration comprises hardware requirements, software environments, and configurations required for efficient experimentation and result evaluation.

3.7.1 HARDWARE SPECIFICATIONS

The experiments were performed on a system with the following specifications:

- System: LAPTOP-V02R5VND, HONOR MagicBook X14
- CPU: Intel(R) Core(TM) i5-10210U CPU @ 1.60GHz 2.11 GHz
- GPU: NVIDIA GeForce RTX 3060, 12GB GDDR6
- RAM: 16 GB DDR4
- Storage: 1 TB NVMe SSD
- Operating System: 64-bit operating system, x64-based processor

These specifications were chosen to ensure sufficient computational power for handling high-dimensional tabular data, image data, and training deep learning models. The integration of GPU acceleration significantly reduced the training times for out-of-distribution models reliant on heavy matrix computations.

3.7.2 SOFTWARE ENVIRONMENT

The experiments were conducted using a sophisticated software framework created to handle the intricacies of image conversion, data processing, and anomaly detection using out-of-distribution (OOD) detection models. The libraries and tools used are described in detail.

3.7.3 PROGRAMMING LANGUAGE

Python 3.10: Selected for its extensive library ecosystem and robust machine learning and data science capabilities.

3.7.4 LIBRARIES

Python packages like TensorFlow, Scikit-learn, and DeepInsight were used to accomplish all of the experiments carried out in this study. To encourage reproducibility, the whole implementation, including model training scripts and image visualisation code, is accessible in a GitHub repository (Hussain, GitHub Repository, 2025).

1. **Data Handling and Preprocessing:**

   Pandas: Specifically designed to handle and analyze tabular data, such as managing numerical and categorical variables, cleaning datasets, and importing CSV files.

   NumPy: Handled the efficient use of arrays and numerical calculations during the preprocessing and analysis of the data.

   Matplotlib: Used to ensure data integrity and validate preprocessing in data visualization and image conversion.

2. **Scikit-learn:**
   Sklearn, also called Scikit-learn, is an open-source Python library for data modeling and machine learning. It has numerous clustering, regression, and classification algorithms available (Pedregosa, 2011).

   To preprocess and use baseline models, this library was required:

   Train_test_split: Used to separate datasets into subsets for testing and training.

   MinMaxScaler and **StandardScaler**: Used to normalise numerical data by feature scaling.

LabelEncoder: Used for converting categorical variables into a format that machine learning systems can use.

RandomForestClassifier: Used to compare image-based out-of-distribution OOD detection methods with a baseline anomaly detection model.

IsolationForestClassifier: Implemented as a robust baseline model for anomaly detection, it is particularly good at identifying anomalies and outliers in tabular data. Additionally, it served as a comparative model for image-based OOD detection methods.

Accuracy_score, precision_score, recall_score, and f1_score: Metrics used to evaluate how well both baseline and advanced models perform.

3. **DeepInsight:**

ImageTransformer: Used to assign pixel positions to features according to their correlations and relevance during the conversion of tabular data into images.

CAMFeatureSelector: Used for detecting important features to enhance the data visualization in the context of the image.

4. **PyOD:**

PyOD offers a wide range of tools for detecting anomalies and outliers, including support for many algorithms. PyOD is frequently used in both industry and academics to find anomalies in complicated and high-dimensional datasets. (Zhao et al., 2019)

We implemented PyOD baseline anomaly detection methods, such as AutoEncoder and Isolation Forest, to assess the performance of image-based models.

5. **TensorFlow/Keras:**

Layers: Variational AutoEncoders (VAE) and other advanced out-of-distribution (OOD) detection models were developed using a variety of layers, including Input, Dense, Flatten, and Reshape.

Models: Neural network frameworks were constructed using the Model and Sequential classes.

Adam Optimizer: To effectively train deep learning models, the Adam Optimizer was used.

6. **Torch (PyTorch):**

Utilities: Neural network layer construction was handled by the nn module, and optimization-related activities were handled by the optim module.

Torchvision.transforms: Image transformations, such as resizing and augmentation, were carried out using this module.

7. **UMAP:** The UMAP provides a better representation and understanding of data distributions by lowering dimensionality and visualizing high-dimensional datasets (McInnes, 2018). It is a great tool for clustering, anomaly detection, and data preprocessing since it is very efficient at maintaining the local and global structure of datasets.
8. **OS**: It offered features for managing files while data processing and doing experiments.

3.7.5 DEVELOPMENT ENVIRONMENT

Jupyter Notebook: The main platform for iterative development, experimentation, and visualization facilitates effective exploration and troubleshooting of the experimental process (Kluyver, 2016).

3.8 COMPARATIVE EVALUATION

The study will compare the results of image-based OOD detection techniques with conventional tabular-based anomaly detection techniques to evaluate:

Effectiveness: evaluating if models based on images are more accurate at identifying anomalies.

Robustness to Different Attack Types: Examining how well each model handles different kinds of attacks, feature distributions, and dataset imbalances.

Scalability and Feasibility: Examining the computing requirements and potential use in large-scale, real-time intrusion detection systems.

4

4 RESULTS & ANALYSIS

The experiments in this study were carried out in the Jupyter Notebook environment with Python as the programming language. The implementation comprises data processing, DeepInsight-based image transformation, and training of both tabular classifiers and ODD models. To ensure reproducibility and transparency, the entire implementation and experiment scripts are made publicly accessible in a GitHub repository (Hussain, GitHub Repository, 2025).

4.1 OVERVIEW OF THE EXPERIMENTAL PIPELINE:

The whole experimental procedure used in this study to assess deep Out-of-Distribution (OOD) intrusion detection techniques and traditional machine learning models is described in this part. A fair and methodical comparison of tabular and image-based approaches under Two-Class and One-Class learning paradigms was made possible by the pipeline's thorough setup.

DeepInsight-based image transformation, model training, multi-metric evaluation, and data processing and feature normalization are the first sequential steps of the experimental system. To guarantee that the research captures behavior in both software-defined networking and IoT contexts, two benchmark datasets, SDN20 and BAIoT_mirai, were utilized.

First, basic tabular network traffic features were cleaned, z-score standardization was used to normalize them, and stratified sampling was used to separate them into training and testing subsets. This guaranteed impartial model evaluation and the preservation of class distribution across splits.

In the second step, high-dimensional tabular features were converted into two-dimensional image representations using the DeepInsight approach. To maintain the neighborhood associations between features, UMAP was employed as the feature embedding technique. Convolutional architectures were able to interpret network traffic in a spatial context due to the generated pixel grids.

In order to achieve robust baseline performance, the third stage required training traditional tabular models (Random Forest and Isolation Forest). These models offer an upper-bound reference for supervised learning and work directly on the normalized feature space.

In the fourth stage, DeepInsight images were used to train deep OOD models, such as Autoencoder (AE), Variational Autoencoder (VAE), and DeepSVDD. Every model was assessed in both One-Class and Two-Class environments to examine how it behaved under different levels of supervision.

Lastly, several evaluation metrics, including training and testing accuracy, were used to assess the model's performance. To ensure the conclusions were reliable, cross-dataset consistency and model stability were also evaluated.

This pipeline's organized architecture makes it possible to thoroughly assess various OOD-based intrusion detection techniques and offers insightful information about their usefulness in real-world scenarios.

4.2 ANALYSIS OF DEEPINSIGHT IMAGE GENERATION:

The DeepInsight transformation is a crucial part of the suggested framework for intrusion detection. Since the deep models used in this study, Autoencoder (AE), Variational Autoencoder (VAE), and DeepSVDD, are developed to take advantage of spatial feature correlations, it is crucial to first assess the features and quality of the generated image representations.

A thorough examination of the DeepInsight procedure, including feature embedding behavior, spatial distribution features, and visual evaluation of the final images, is given in this part. Because the quality of the tabular-to-image transformation greatly influences the efficacy of convolutional OOD models, it is critical to comprehend these features.

4.2.1 DEVELOPMENT ENVIRONMENT:

The DeepInsight pipeline mapped high-dimensional tabular features into a two-dimensional space using Uniform Manifold Approximation and Projection (UMAP). This embedding step's main goal is to maintain the local neighborhood structure of features so that linked features are placed in proximity to one another in the resulting image grid.

Due to its superior computational efficiency and capacity to preserve both local and global data structure, UMAP was chosen over other dimensionality reduction methods like t-SNE. Both the SDN20 and BAIoT_mirai datasets showed stable and well-separated feature architectures during the embedding process.

Additionally, in the One-Class context, the embedding procedure was carried out using only normal training data. This design decision increases the sensitivity of downstream anomaly detection models to deviations from normal patterns by ensuring that the spatial architecture reflects the fundamental structure of normal network traffic.

Visualization of UMAP feature embedding:

The UMAP-based dimensionality reduction method used to create the spatial feature layout for DeepInsight transformation is demonstrated in the code sample that follows:

UMAP Python Code:

```
reducer = umap.UMAP(
    n_components=2,
    min_dist=0.1,
    metric='cosine',
    n_jobs=-1,
    random_state=42

)

pixel_size = (8, 8)
it = ImageTransformer(
    feature_extractor=reducer,
    #feature_extractor='pca',
    pixels=pixel_size)
```

The two-dimensional UMAP projection of the high-dimensional tabular features is displayed in Figures 1 and 2. The efficacy of the DeepInsight spatial encoding technique is supported by the observation that associated features create localized clusters.

```
it.fit(X_train, plot=True)
```

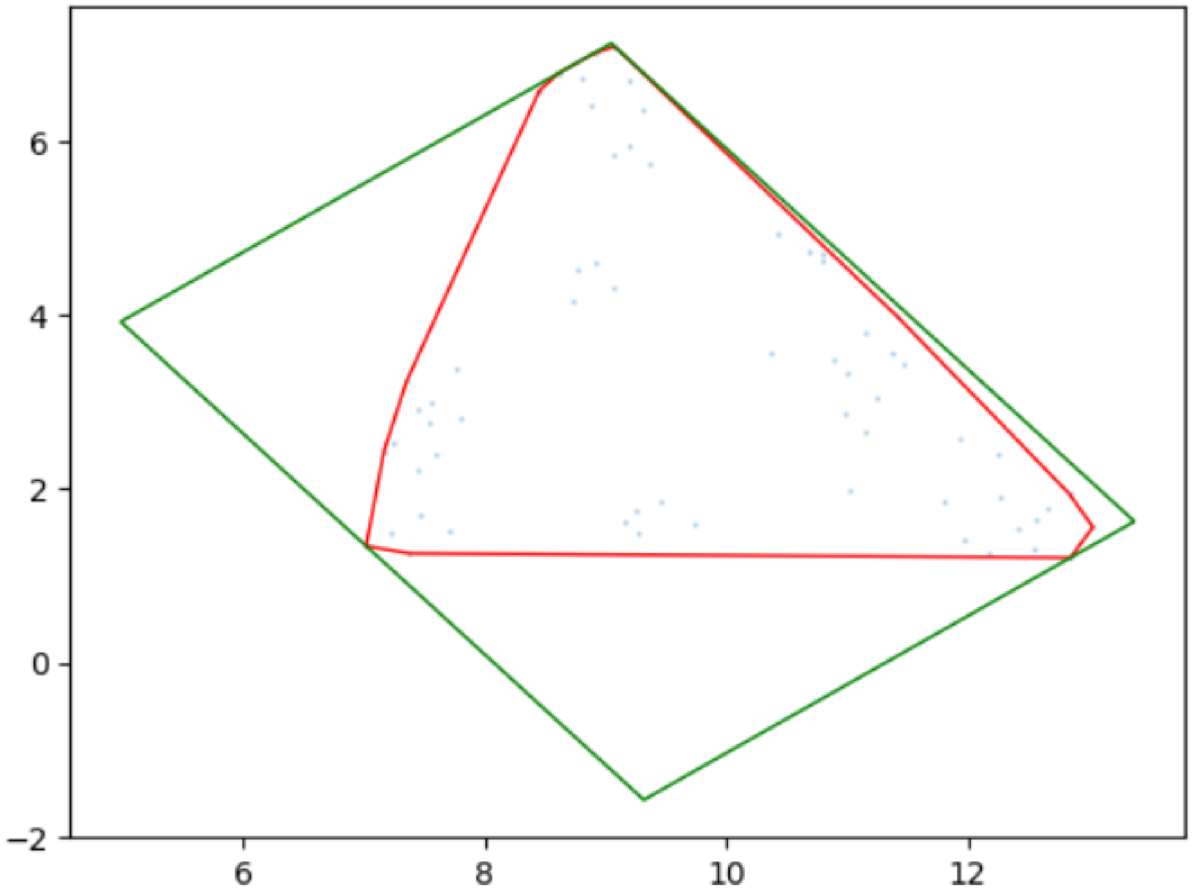

FIGURE 1 (TRAINING): UMAP PROJECTION OF TABULAR FEATURES IN 2-DIMENSION FOR DEEPINSIGHT IMAGE CONSTRUCTION.

```
it.fit(X_test, plot=True)
```

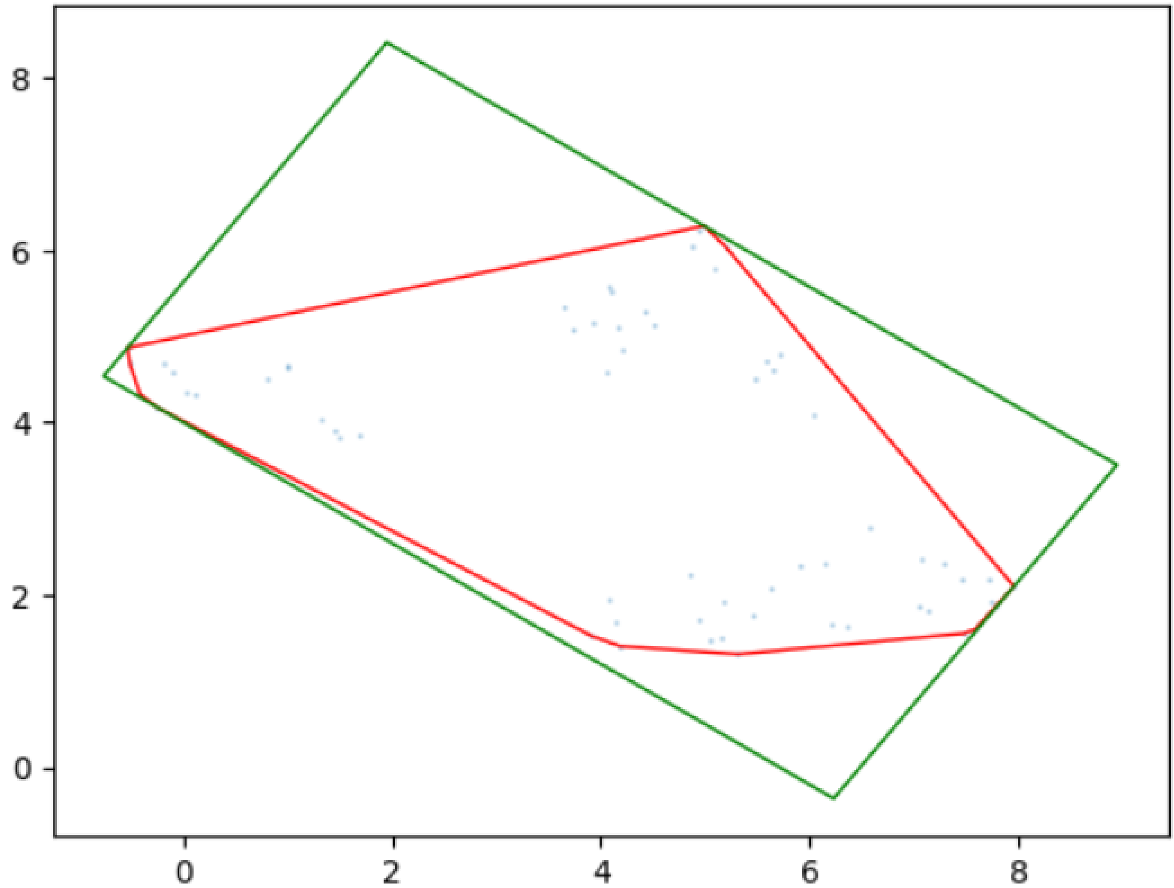

FIGURE 2 (TESTING): UMAP PROJECTION OF TABULAR FEATURES IN 2-DIMENSION FOR DEEPINSIGHT IMAGE CONSTRUCTION.

4.2.2 CHARACTERISTICS OF THE GENERATED IMAGE:

To construct visual representations, tabular samples were mapped into a fixed pixel grid of size 8 × 8 after the UMAP embedding. To provide numerical stability during neural network training, each feature value was allocated to its appropriate spatial location, and pixel intensities were normalized to the [0,1] range.

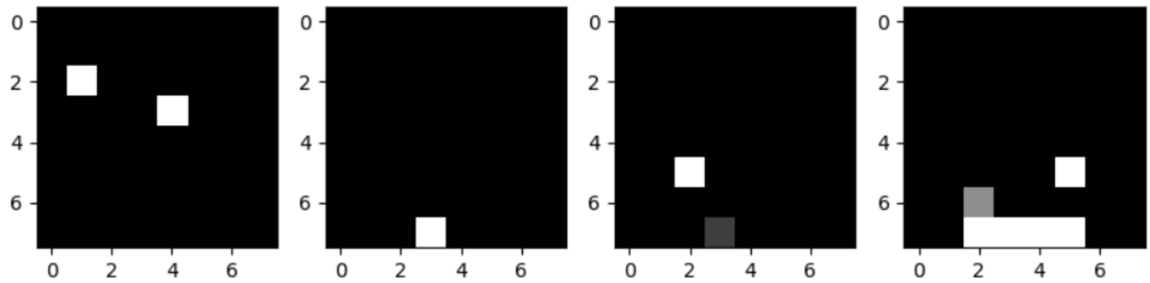

FIGURE 3 (TRAINING): IMAGE SAMPLES CREATED BY DEEPINSIGHT FROM THE TRAINING DATASET.

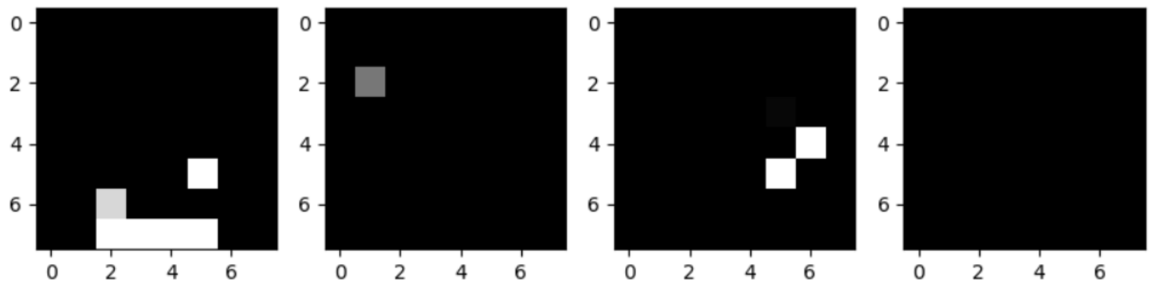

FIGURE 4 (TESTING): IMAGE SAMPLES CREATED BY DEEPINSIGHT FROM THE TESTING DATASET.

The transformation process starts with projecting high-dimensional features into a two-dimensional space using dimensionality reduction techniques like UMAP. Each feature is assigned a specific location on the image grid based on its spatial position in the reduced feature space. The feature values are then transformed into pixel intensities in the resulting images.

Figures 4 and 3 show multiple DeepInsight-generated images from the training and testing datasets. Each image represents a single network traffic event. Each pixel's greyscale intensity indicates the normalised value of its related network feature.

Bright pixels indicate greater feature values, whereas darker pixels indicate lower feature values. The spatial arrangement of these pixels allows convolutional neural networks to detect local correlations between features, allowing the model to learn meaningful patterns from network traffic data. This representation effectively allows deep learning models like DeepSVDD, Autoencoder, and Variational Autoencoder to analyse tabular intrusion detection data using convolutional architectures.

Moreover, Several significant findings about the produced pictures were noted:

First, instead of random noise, the images showed organized spatial patterns. Similar intensity distributions were seen in regions that corresponded to associated features, suggesting that DeepInsight was successful in capturing significant correlations between features.

Secondly, solid gradient-based optimization depends on the normalization method producing constant dynamic ranges across samples. Convolutional models may suffer from vanishing or exploding gradients if they are not properly normalized.

Third, the 8 x 8 image resolution offered a fair compromise between computing performance and spatial detail. Significantly bigger grids increased computing cost without appreciable performance advantages, whereas preliminary trials with smaller grids produced excessive feature overlap.

Furthermore, visual representation showed that anomalous traffic samples showed inconsistent intensity distributions, while normal traffic samples generated comparatively smooth and consistent spatial patterns. The ability of reconstruction-based models to successfully identify anomalies in subsequent experiments can be clearly explained by this finding.

4.3 PERFORMANCE OF THE BASELINE TABULAR MODEL:

It is crucial to establish a solid baseline performance using traditional tabular machine learning techniques before evaluating the suggested Deep Insight-based deep learning models. In addition to offering as an upper-bound reference, baseline models enable assessing whether the additional complexity of deep OOD techniques is justified.

This study evaluated two popular tabular models:

- Random Forest (supervised)
- Isolation Forest (unsupervised)

Without using the DeepInsight transformation, both models were trained directly on the normalized tabular feature space. Accuracy, precision, recall, and F1-score were used to evaluate performance on the SDN20 and BAIoT_mirai datasets.

Moreover, the code of both models is available in the project repository, which contains the implementation of both Random Forest and Isolation Forest, including preprocessing and evaluation of these models (Hussain, GitHub Repository, 2025).

4.3.1 PERFORMANCE OF RANDOM FORESTS:

Random Forest's demonstrated efficacy in intrusion detection literature and its resilience in managing high-dimensional structured data led to its selection as the main supervised baseline.

4.3.1.1 SDN20 RANDOM FOREST RESULTS:

Random Forest showed remarkably good classification performance on the SDN20 dataset. As indicated by Table 4.3.1.1, the model's testing accuracy was 0.99815, and its training accuracy was 0.99965.

The testing set's precision and recall values stayed above 0.9979, yielding an F1-score of 0.99804. The model does not show substantial overfitting and generalizes well, as evidenced by the incredibly small difference between training and testing measures.

This result implies that, when used by ensemble tree models, the statistical flow features in SDN20 are highly discriminative from a behavioral perspective. A major factor in Random Forest's efficiency is its capacity to represent nonlinear feature interactions and carry out implicit feature selection.

Additionally, the near-perfect recall shows that most attack events were correctly identified, which is crucial in cybersecurity situations where missed detections might have catastrophic consequences.

4.3.1.1 RESULTS OF RANDOM FOREST ON BAIOT_MIRAI

On the BAIoT_mirai dataset, the Random Forest classifier performed flawlessly, achieving 100% accuracy, precision, recall, and F1-score on both training and testing sets.

This outcome shows that in the evaluated feature space, the model was able to fully distinguish between abnormal and normal traffic. Such performance shows how well supervised learning works, but it also implies that the dataset has good class separability.

These findings provide a very solid supervised benchmark from an experimental standpoint. It is important to properly understand this result, though. The assumption of completely labelled and separable attack data may not hold true in real-world intrusion detection environments, especially in environments at risk from zero-day threats.

Consequently, Random Forest does not adequately address the open-world issues that drive the usage of OOD and One-Class techniques, even if it offers a crucial upper-bound reference.

4.3.2 PERFORMANCE OF ISOLATION FOREST:

Isolation Forest was assessed as a baseline for unsupervised anomaly detection. Isolation Forest, in contrast to Random Forest, finds anomalies based on how simple it is to isolate samples using random partitioning rather than requiring labelled attack data.

4.3.2.1 RESULTS OF ISOLATION FOREST ON SDN20

Isolation Forest obtained training accuracy of 0.8999 and a testing accuracy of almost 0.9000, and a F1-score of 0.94737 on the SDN20 dataset. This performance is noteworthy considering the algorithm's unsupervised nature, even though it is lower than Random Forest.

The model attained flawless precision (1.0), which is a particularly interesting finding. This suggests that the model is very dependable when it identifies a sample as abnormal. The recall number (~0.90), however, indicates that some attack instances might go unnoticed.

This behavior is in line with Isolation Forest's design. The technique may have trouble capturing complex or subtle attack patterns seen in high-dimensional network traffic since it depends on random partitioning rather than deep feature representation.

However, the model shows that unsupervised detection may capture a significant amount of malicious behavior and offers a helpful, lightweight baseline.

4.3.2.2 RESULTS OF THE ISOLATION FOREST ON BAIOT_MIRAI

The BAIoT_mirai dataset showed a comparable performance pattern. Isolation Forest obtained a nearly 0.9039 of training accuracy and testing accuracy of roughly 0.90 and an F1-score near 0.947 and a testing accuracy of roughly 0.90.

Isolation Forest offers a stable but moderate detection of anomaly capacity, according to the consistency across datasets. Although the method is good at spotting clear anomalies, it is not as good as deep reconstruction-based methods or supervised models.

In practical terms, Isolation Forest's computational efficiency may make it valuable in environments with limited resources. More expressive models, however, might be better for high-security applications that demand maximal detection coverage.

4.3.3 COMPARISON OF BASELINE MODELS:

Several significant insights are revealed by comparing the baseline models.

First, in every evaluated metric, supervised Random Forest performs noticeably better than Isolation Forest. This demonstrates that having access to labelled attack data is a major benefit for accurately determining decision boundaries.

Second, Isolation Forest's consistent but moderate performance shows that unsupervised techniques may still identify significant anomaly patterns, but with less sensitivity than supervised learning.

Lastly, these baseline results demonstrate the primary motivating factor behind this thesis: real-world intrusion detection systems need to be resilient against previously unknown attacks, even while supervised models perform exceptionally well in closed-world environments.

Above all, the baseline experiments demonstrate that Random Forest is still a very successful supervised classifier for structured intrusion detection data, with near-perfect results on the SDN20 and BAIoT_mirai datasets. Although Isolation Forest offers an appropriate unsupervised baseline. This motivates the thorough analysis of deep OOD models; these results provide a solid benchmark for assessing the DeepInsight-based deep OOD models in the sections that follow.

TABLE 1: RANDOM FOREST PERFORMANCE

Dataset		Accuracy	Precision	Recall	F1-Measure
	Training	0.999	0.999	0.999	0.999
SDN20	**Testing**	0.998	0.997	0.998	0.998
	Training	1.000	1.000	1.000	1.000
BAIoT_mirai	**Testing**	1.000	1.000	1.000	1.000

TABLE 2: ISOLATION FOREST PERFORMANCE

Dataset		Accuracy	Precision	Recall	F1-Measure
	Training	0.899	1.000	0.899	0.947
SDN20	**Testing**	0.900	1.000	0.900	0.947
	Training	0.903	1.000	0.903	0.949
BAIoT_mirai	**Testing**	0.900	1.000	0.949	0.947

The baseline machine learning classifiers' performance was evaluated using the SDN20 and BAIoT datasets. Figure:03 compares the training and testing accuracy of Random Forest and Isolation Forest models, respectively. Tables: 1 and 2 show extensive numerical data for training and testing accuracy, while Figure 3 presents a visual comparison of the models.

As we can see from Table 1, Random Forest showed extraordinarily high accuracy on both datasets. On the SDN20 dataset, it has a training accuracy of 0.99965 and a testing accuracy of 0.99815. On the BAIoT dataset, the model still obtained perfect classification performance, with 1.0 training and testing accuracy. In comparison, Isolation Forest performed moderately, with testing accuracy around 0.90 on both datasets, as we can see in Table: 2.

The comparison is presented in Figure: 03 clearly demonstrates the difference in performance between the two approaches across both datasets. And, these findings show that tabular supervised classifiers like random forest can achieve extremely high detection accuracy when labelled data is available.

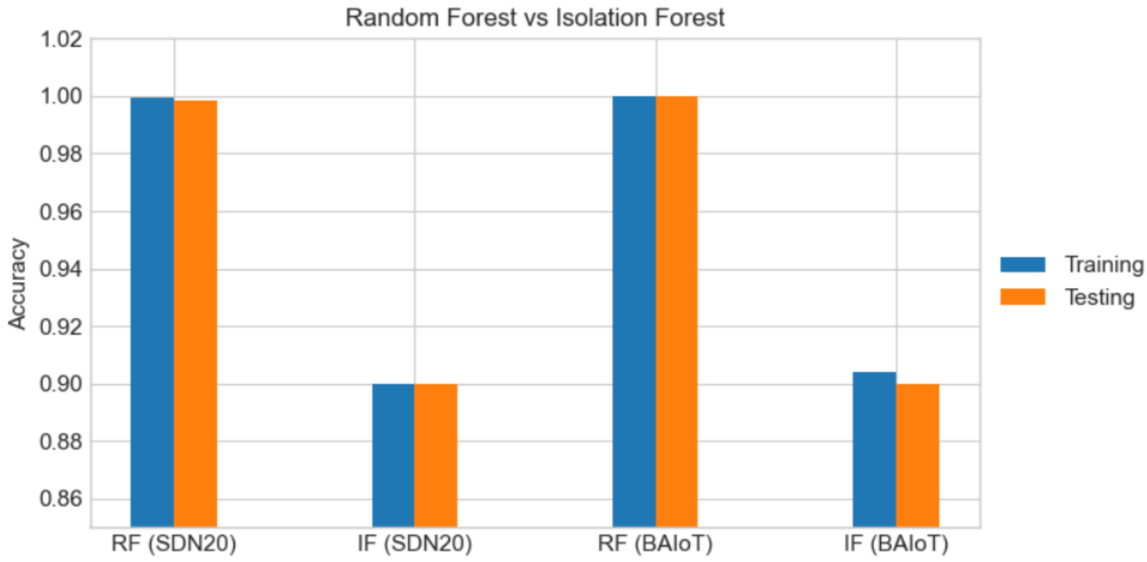

FIGURE 5: *COMPARISON OF RANDOM FOREST AND ISOLATION FOREST PERFORMANCE ACROSS THE SDN20 AND BAIOT DATASETS.*

4.4 EVALUATION OF OOD TECHNIQUES FOR INTRUSION DETECTION WITH TWO-CLASS SETTINGS:

The next stage of the experiment analyzes deep learning-based Out-of-Distribution (OOD) models in the Two-class Intrusion Detection environment after consistent tabular baselines have been established. In this setup, the DeepInsight pipeline was used to convert tabular data into image representations before. Then, models were trained using both normal and attack samples.

This section's main goal is to evaluate how well deep convolutional architectures can simulate structured network traffic when spatial interaction between features is added.

Three ODD models were evaluated:

1. Deep Support Vector Data Description (DeepSVDD)
2. Autoencoder (AE)
3. Variational Autoencoder (VAE)

TensorFlow and Keras were used to implement the training and evaluation processes for the DeepSVDD, Autoencoder, and Variational Autoencoder models. The public project repository (Hussain, GitHub Repository, 2025) contains the full implementation scripts that were used to generate these results.

4.4.1 PERFORMANCE OF DEEPSVDD WITH TWO-CLASS INTRUSION DETECTION:

4.4.1.1 RESULTS OF DEEPSVDD ON SDN20 WITH TWO-CLASS INTRUSION DETECTION:

On the SDN20 dataset, DeepSVDD's test accuracy in the supervised setting was 0.9811, and its training accuracy was 0.9809. Stable learning behavior and reasonable generalization are indicated by the very modest difference between training and testing performance.

DeepSVDD aims to learn a compact hypersphere around normal samples in the learned feature space from a modelling standpoint. The model gains from a better representation shape when supervised labels are provided, allowing for the comparatively good performance seen.

However, DeepSVDD's accuracy is marginally lower than that of reconstruction-based models like AE or VAE. This implies that when full label supervision is available, reconstruction-based methods may be more expressive than hypersphere-based representation learning.

In spite of this limitation, the model continues to show competitive performance and validates that DeepSVDD can function well on DeepInsight-transformed network traffic.

4.4.1.2 RESULTS OF DEEPSVDD ON BAIOT_MIRAI WITH TWO-CLASS SETTINGS:

DeepSVDD obtained a training accuracy of 0.985 and a testing accuracy of 0.9848 on the BAIoT_mirai dataset. The model generalizes fairly well across different network environments based on the consistency between the two datasets.

The significantly greater performance in comparison to SDN20 could be explained by variations in the feature distributions and traffic patterns included in SDN20 data. However, the findings show that, in supervised environments, DeepSVDD is still a practical deep anomaly detection method.

Overall, before moving on to the more difficult One-Class environment, the supervised DeepSVDD trials provide an insightful performance reference.

4.4.2 PERFORMANCE OF AUTOENCODER WITH TWO-CLASS INTRUSION DETECTION:

4.4.2.1 RESULTS OF AUTOENCODER ON SDN20 WITH TWO-CLASS SETTINGS:

In the supervised environment, the Autoencoder performed exceptionally well. The model obtained a training accuracy of 0.999348 and a testing accuracy of 0.999318 on the SDN20 dataset.

This almost flawless result shows that the Autoencoder was successful in capturing the DeepInsight-generated images' underlying structure.

The Autoencoder's reconstruction-based learning mechanism is responsible for its excellent performance. The model implicitly captures discriminative spatial features found in the DeepInsight images by learning to accurately reproduce attack and normal patterns.

Additionally, it appears that the convolutional encoder/decoder architecture is a good fit for simulating the spatial correlations that the UMAP-based feature embedding introduces.

4.4.2.2 RESULTS OF AUTOENCODER ON BAIOT_MIRAI WITH TWO-CLASS SETTINGS:

The BAIoT_mirai dataset exhibited a similarly strong performance trend. With a testing accuracy of 0.999734 and training accuracy of 0.999767, the Autoencoder is equivalent to the supervised Random Forest baseline.

The reconstruction-based method generalizes effectively across diverse network environments, as evidenced by its cross-dataset consistency. The outcomes provide further evidence of Deep Insight's effectiveness in enabling the deployment of deep convolutional models on structured intrusion data.

In practical terms, one of the most competitive deep models in the supervised scenario is the Autoencoder.

4.4.3 PERFORMANCE OF VARIATIONAL AUTOENCODER WITH TWO-CLASS INTRUSION DETECTION

4.4.3.1 RESULTS OF VARIATIONAL AUTOENCODER ON SDN20 WITH TWO-CLASS SETTINGS

With a testing accuracy of 0.9952 and a training accuracy of 0.995, the Variational Autoencoder similarly showed impressive performance on the SDN20 dataset.

The VAE still produces very competitive results, while being slightly lower than the conventional Autoencoder. The KL-divergence regularization factor in the VAE loss function, to encourage a smoother latent space at the cost of slightly less reconstruction precision, explains the marginal performance difference.

The VAE exhibits good generalization behavior and maintains an effective anomaly modelling capability in spite of this trade-off.

4.4.3.2 RESULTS OF VARIATIONAL AUTOENCODER ON BAIOT_MIRAI WITH TWO-CLASS SETTINGS:

The VAE obtained a testing accuracy of 0.9949 on the BAIoT_mirai dataset, which is once more marginally less than the Autoencoder but still extremely good.

Also, Probabilistic latent modelling does not substantially decrease supervised classification capability, as evidenced by the consistently strong performance across both datasets. Rather, it offers a more organized latent representation, which could be especially useful in an One-Class environment.

4.4.4 COMPARATIVE ANALYSIS OF ODD MODELS FOR INTRUSION DETECTION IN A TWO-CLASS ENVIRONMENT:

A comparison analysis of the supervised deep models identifies a number of significant patterns.

First, in the Two-Class environment, reconstruction-based models (AE and VAE) consistently surpass DeepSVDD. This implies that, in contrast to hypersphere-based embedding, direct reconstruction objectives could be able to capture richer spatial information from DeepInsight images.

Second, among the deep ODD models, the Autoencoder and Variational Autoencoder perform best overall; both models surpassed the Random Forest tabular classifier. This suggests that when label supervision is available, convolutional reconstruction networks are quite successful for structured intrusion detection.

Third, even though DeepSVDD's accuracy is marginally reduced, its performance is still competitive and consistent across datasets, demonstrating its potential as a deep anomaly detection technique.

Lastly, the consistent behavior between SDN20 and BAIoT_mirai supports the generalizability of the experimental results and strengthens the validity of the observed trends. Above all, the supervised deep OOD experiments show that convolutional models enabled by DeepInsight may achieve very high intrusion detection performance. The Autoencoder achieves the best supervised outcomes among the assessed techniques, closely followed by the Variational Autoencoder; DeepSVDD performs moderately but consistently.

TABLE 3: ODD TECHNIQUES IN TWO-CLASS SETTING PERFORMANCE

Dataset	Model		Accuracy	Precision	Recall	F1-Measure
		Training	0.981	1.000	0.962	0.980
	DeepSVDD	**Testing**	0.980	1.000	0.961	0.980
		Training	0.999	0.999	0.999	0.999
SDN20	**AE**	**Testing**	0.999	0.999	0.999	0.999
		Training	0.995	0.990	1.000	0.995
	VAE	**Testing**	0.995	0.990	1.000	0.995
		Training	0.985	0.970	1.000	0.985
	DeepSVDD	**Testing**	0.984	0.970	1.000	0.985
		Training	0.999	1.000	0.999	0.999
BAIoT_mirai	**AE**	**Testing**	0.999	1.000	0.999	0.999
		Training	0.995	0.990	1.000	0.995
	VAE	**Testing**	0.994	0.989	1.000	0.994

Figure 4 shows how DeepSVDD, Autoencoder, and Variational Autoencoder perform in a Two-Class setting.

According to Table 4, the Autoencoder has the highest testing accuracy of the models evaluated, scoring 0.999318 on the SDN20 dataset and 0.999734 on the BAIoT dataset. The Variational Autoencoder also performed well, with testing accuracy scores of 0.9952 and 0.9949 for the SDN20 and BAIoT datasets, respectively.

DeepSVDD had significantly lower testing accuracy than reconstruction-based models, with 0.9809 on SDN20 and 0.9848 on the BAIoT dataset; it still performed well.

As shown in Figure 04, the Autoencoder and Variational Autoencoder models outperform DeepSVDD in the Two-Class classification scenario, demonstrating the efficiency of reconstruction-based learning for capturing network traffic patterns.

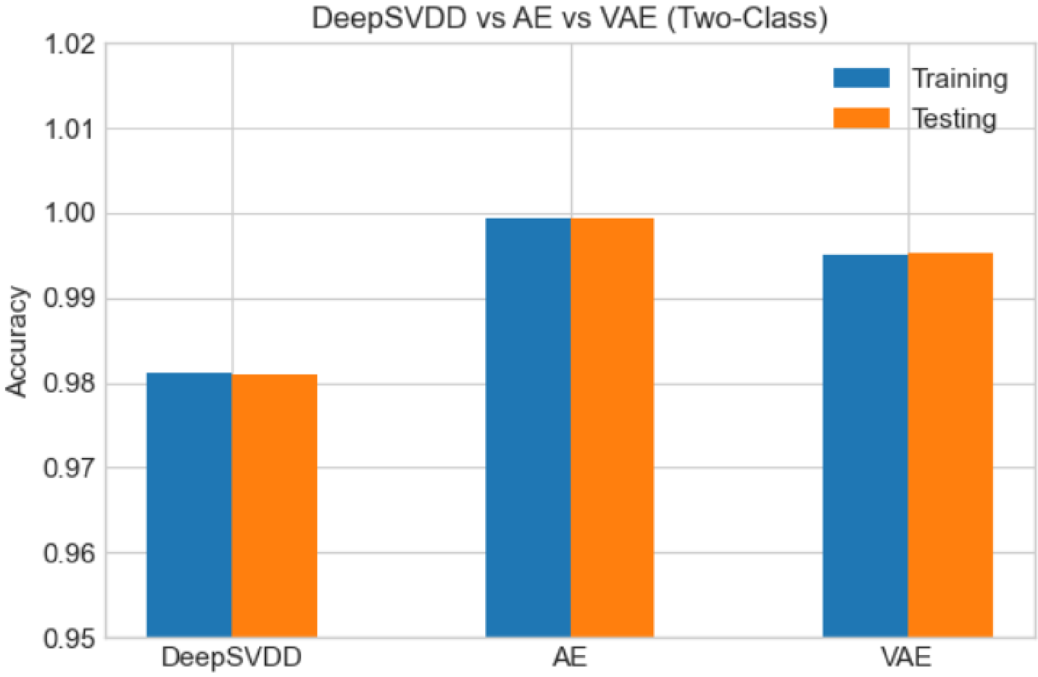

FIGURE 6: *COMPARES THE PERFORMANCE OF DEEPSVDD, AUTOENCODER, AND VARIATIONAL AUTOENCODER IN A TWO-CLASS ENVIRONMENT.*

4.5 EVALUATION OF OOD TECHNIQUES FOR INTRUSION DETECTION WITH ONE-CLASS SETTINGS:

While real-world intrusion detection systems frequently function in environments where malicious patterns are incomplete, changing, or completely unknown, supervised models offer good performance when labelled attack data is available. One-Class anomaly detection is, therefore, a more practical and difficult evaluation case.

This section evaluated the deep OOD models' ability to identify previously unknown attacks after they were trained solely on normal traffic samples. This configuration puts the suggested DeepInsight-based framework's robustness and applicability to the test.

Three ODD models were evaluated:

1. Deep Support Vector Data Description (DeepSVDD)
2. Autoencoder (AE)
3. Variational Autoencoder (VAE)

The main metric used to assess performance was testing accuracy on mixed normal and attack traffic. You may find the whole implementation and experiment configuration scripts in the project repository (Hussain, GitHub Repository, 2025).

4.5.1 PERFORMANCE OF DEEPSVDD WITH ONE-CLASS INTRUSION DETECTION:

4.5.1.1 RESULTS OF DEEPSVDD ON SDN20 WITH ONE-CLASS INTRUSION DETECTION:

On the SDN20 dataset, DeepSVDD's testing accuracy under the One-Class environment was 0.8252, whereas its training accuracy on normal-only data was roughly 0.9532.

The training approach and threshold tuning effectively brought the hypersphere learning process stabilized. DeepSVDD aims to learn a compact border around normal samples in the latent space from a modelling perspective. Samples outside of this learnt hypersphere are identified as anomalies during testing.

The model performs substantially lower than reconstruction-based methods, yet shows a respectable detection capability.

This suggests that the complex variability found in DeepInsight-transformed network data may be difficult for hypersphere-based representation to fully capture in an One-Class environment.

4.5.1.2 RESULTS OF DEEPSVDD ON BAIOT_MIRAI WITH ONE-CLASS INTRUSION DETECTION:

With a training accuracy of 0.947114 and a testing accuracy of 0.959489, DeepSVDD performed noticeably better on the BAIoT Mirai dataset. This improvement implies that, in comparison to the SDN20 dataset, the DeepSVDD model was more successful in learning the normal traffic distribution within the BAIoT dataset.

However, when applied to highly complex network traffic distributions, the performance level around the 90-95% range points to an underlying restriction of the hypersphere formulation in One-Class settings.

In practical terms, DeepSVDD can offer a moderate level of anomaly detection abilities, but to compete with reconstruction-based techniques, it might need careful hyperparameter tuning or architectural enhancements.

4.5.2 PERFORMANCE OF AUTOENCODER WITH ONE-CLASS INTRUSION DETECTION:

4.5.2.1 RESULTS OF AUTOENCODER ON SDN20 WITH ONE-CLASS INTRUSION DETECTION:

In the One-Class environment, the Autoencoder showed excellent robustness. The model obtained a training accuracy of 0.948231 on normal samples and a testing accuracy of 0.96078 on the SDN20 dataset.

This performance shows that the Autoencoder was able to identify abnormal deviations through reconstruction error after successfully learning the underlying distribution of normal network data.

The Autoencoder models the entire data distribution, in contrast to DeepSVDD, which depends on a geometric boundary. Because of this, it is frequently more capable of identifying small differences in complex feature spaces.

The efficiency of this approach was further supported by visual inspection of reconstruction errors, which revealed that anomalous samples created significantly more reconstruction loss than normal traffic.

4.5.2.2 RESULTS OF AUTOENCODER ON BAIOT_MIRAI WITH ONE-CLASS INTRUSION DETECTION:

The Autoencoder obtained a training accuracy of 0.948407 on normal samples and a testing accuracy of 0.999025. The Autoencoder outperformed all other models in terms of One-Class performance on the BAIoT_mirai dataset.

The model continues to show excellent anomaly detection accuracy, demonstrating the benefit of reconstruction-based learning for modelling normal network behavior. The more complex and erratic traffic patterns in SDN20 could be the cause of the performance decline.

Above all, the accuracy demonstrates the Autoencoder's capacity to effectively reconstruct typical traffic while identifying anomalous traffic patterns in the dataset.

These findings again show that reconstruction-based methods, like Autoencoders, work quite well for tasks involving network anomaly detection.

4.5.3 PERFORMANCE OF VARIATIONAL AUTOENCODER WITH ONE-CLASS INTRUSION DETECTION:

4.5.3.1 RESULTS OF VARIATIONAL AUTOENCODER ON SDN20 WITH ONE-CLASS INTRUSION DETECTION:

In both datasets, the Variational Autoencoder similarly demonstrated excellent performance. With a training accuracy of 0.97 and a testing accuracy of 0.9855 on the SDN20 dataset, the VAE has the greatest accuracy of all the models that were evaluated for this dataset.

This shows that the probabilistic latent space that the VAE learned has led to this better performance. The VAE creates a smoother and more realistic representation of normal traffic patterns by using KL-divergence regularization to enforce a structured distribution. Because of this, anomalous data that deviates from this learned distribution result in significantly greater reconstruction errors, which improves the sensitivity of detection.

The positive aspect of probabilistic generative modelling for intrusion detection in open-world environments is demonstrated by the VAE's robust performance.

4.5.3.2 RESULTS OF VARIATIONAL AUTOENCODER ON BAIOT_MIRAI WITH ONE-CLASS INTRUSION DETECTION:

The VAE demonstrated good anomaly detection capabilities on the BAIoT Mirai dataset, with training accuracy of 0.95 and testing accuracy of 0.9737.

The VAE appears to generalize effectively across various network environments, as evidenced by the relatively small performance decline when compared to SDN20, but it was still very competitive.

4.5.4 COMPARATIVE ANALYSIS OF ODD MODELS FOR INTRUSION DETECTION IN ONE-CLASS ENVIRONMENT:

A thorough analysis of the One-Class experiments shows a distinct hierarchy of performance.

Across both datasets, the Variational Autoencoder consistently produced the best results, followed by the Autoencoder. DeepSVDD demonstrated fair but better performance.

Several important conclusions can be drawn:

First, reconstruction-based models are more robust in modelling normal network behavior. These models can show abnormal deviations because they reproduce normal patterns.

Second, the VAE's probabilistic latent structure allows for further generalization capacity, which explains its persistent high performance.

Third, while DeepSVDD improved greatly after training stabilization, its hypersphere formulation is less adaptable to capturing the full range of DeepInsight-transformed traffic.

Finally, the consistent trends seen in both SDN20 and BAIoT_mirai support the veracity of these conclusions.

Summary of Results:

In conclusion, the One-Class evaluation demonstrates that deep OOD models have significant capability for detecting previously unknown network attacks. Among these techniques, the Variational Autoencoder performs the best and most consistently, followed by the Autoencoder, and DeepSVDD produces moderate but reliable results.

These results significantly support the use of reconstruction-based deep models in practical intrusion detection contexts where labelled attack data is restricted or unavailable.

TABLE 4: *ODD TECHNIQUES IN ONE-CLASS SETTINGS PERFORMANCE*

Dataset	Model	Accuracy	Precision	Recall	F1-Measure
	DeepSVDD	0.825	0.968	0.762	0.853
SDN20	**AE**	0.960	0.965	0.753	0.846
	VAE	0.985	0.971	1.000	0.985
	DeepSVDD	0.947	0.905	0.983	0.943
BAIoT_mirai	**AE**	0.999	0.908	1.000	0.951
	VAE	0.973	0.950	1.000	0.974

The findings of the One-Class anomaly detection studies are shown in Table 4 above, while Figure 5 presents a visual comparison of model performance below.

In this case, the models were trained with only normal samples and tested on both normal and attack samples. As shown in Table 4, the Variational Autoencoder has the highest testing accuracy of the models tested, scoring 0.9855 on the SDN20 dataset and 0.9737 on the BAIoT dataset, slightly lower than AE.

Similarly, the Autoencoder performed best, followed by the Variational Autoencoder, with testing accuracy scores of 0.96078 on SDN20 and 0.999025 on BAIoT, where AE surpassed VAE. In comparison, DeepSVDD performed moderately, with testing accuracy scores of 0.825242 and 0.947114 on the SDN20 and BAIoT datasets, respectively.

Figure 4 clearly illustrates the performance differences between different models below, where the reconstruction-based models, like Autoencoder or Variational Autoencoder, exhibit stronger anomaly detection capabilities.

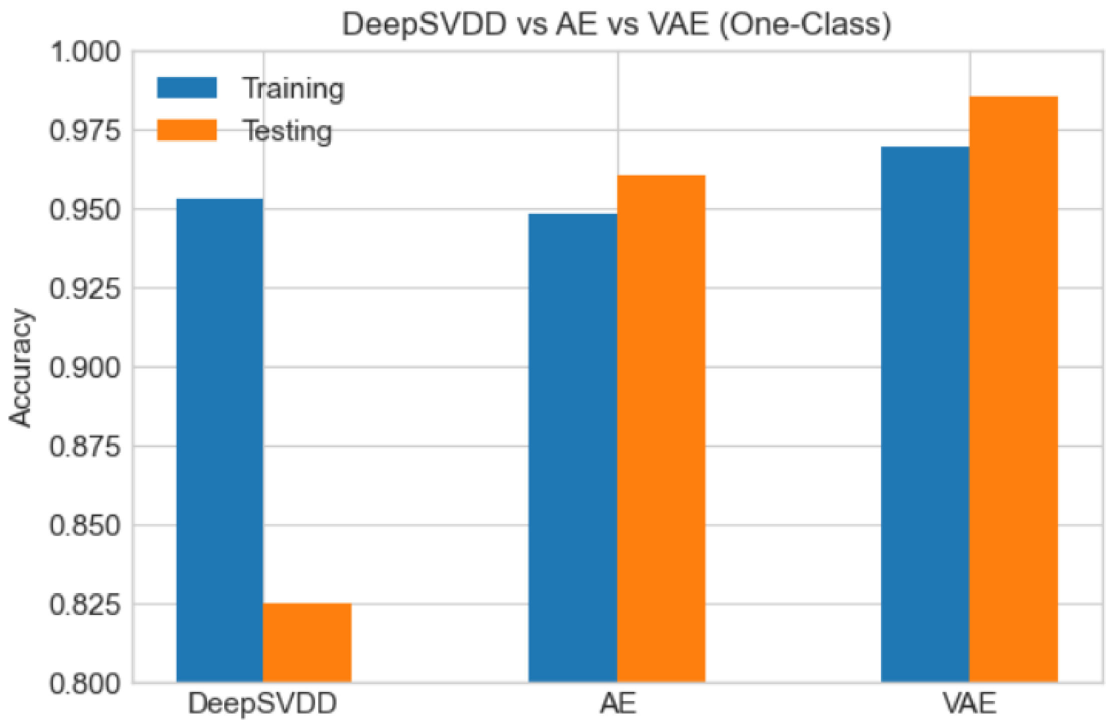

FIGURE 7: ***COMPARES THE PERFORMANCE OF DEEPSVDD, AUTOENCODER, AND VARIATIONAL AUTOENCODER IN A ONE-CLASS ENVIRONMENT.***

4.5.5 COMPARATIVE ANALYSIS OF TABULAR CLASSIFIERS VS ODD MODELS:

Figure 6 shows a complete comparison of tabular classifiers and deep learning-based OOD approaches, including DeepSVDD, Autoencoder, and Variational Autoencoder.

Random Forest achieved the best supervised performance, with test accuracy exceeding 99% on both datasets. Isolation Forest achieved an accuracy of approximately 90%, highlighting the limitations of unsupervised anomaly detection approaches.

Autoencoder and Variational Autoencoder performed similarly to supervised classifiers, specifically in the Two-Class setting.

As seen in Figure 06, reconstruction-based models offer significant intrusion detection capabilities while also supporting anomaly detection scenarios in which labelled attack data is limited.

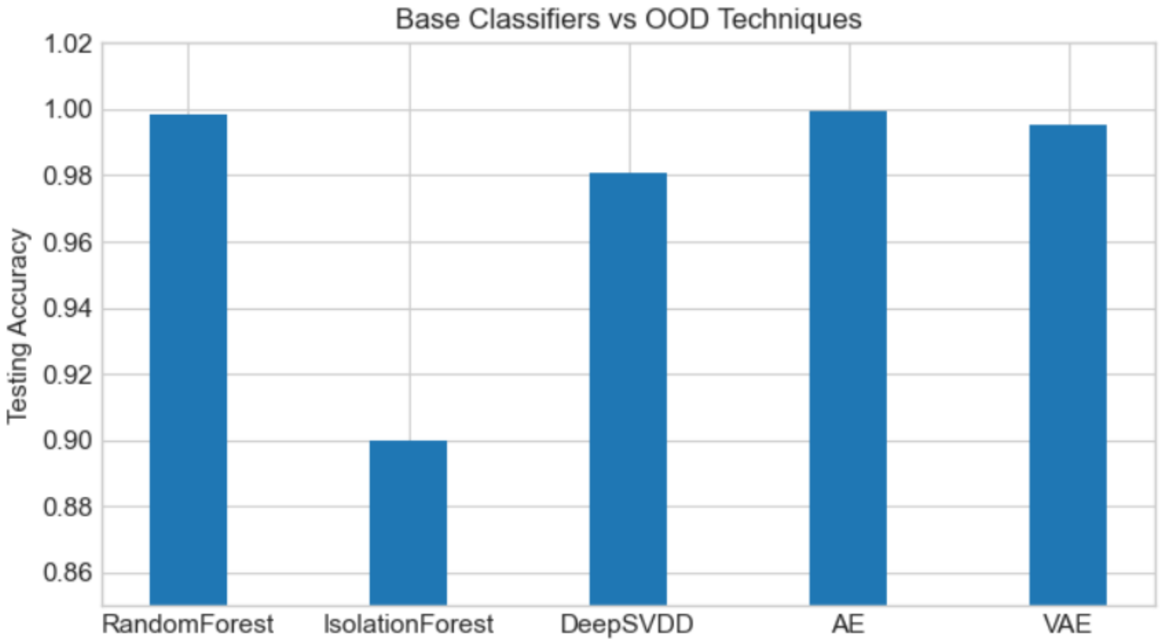

FIGURE 8: ***COMPARES THE PERFORMANCE OF TABULAR CLASSIFIERS AND* OOD APPROACHES.**

Note: The complete experimental implementation used to generate the results given in this chapter has been made publicly available in the project repository for reproducibility and future research (Hussain, GitHub Repository, 2025).

5

5. DISCUSSION & CONCLUSION

5.1 OVERVIEW OF FINDINGS:

This chapter investigated the experimental outcomes of traditional machine learning models and deep learning-based out-of-distribution strategies for intrusion detection on the SDN20 and BAIoT Mirai datasets. The models were evaluated in both supervised (two-class) and One-Class experimental conditions.

The following subsections summarise the most significant findings from the experimental evaluation.

5.2 ANALYSIS OF TABULAR MACHINE LEARNING MODELS:

The outcomes of traditional machine learning models demonstrate the continuous relevance of supervised learning techniques for problems utilising structured network traffic data in intrusion detection. Because Random Forest is an ensemble model and can efficiently handle high-dimensional tabular data, it exhibits a remarkable capacity to capture complicated feature relationships.

Random Forest performs better because it can model non-linear correlations in the data and is resistant to overfitting. Because of this, it is especially well-suited for intrusion detection situations involving complicated and varied network traffic patterns. Additionally, its steady performance on both datasets shows that it has a great capacity for supervised generalisation.

On the other hand, Isolation Forest's comparatively poorer performance illustrates the inherent drawbacks of strictly unsupervised methods. Isolation Forest may have trouble differentiating between small differences in legitimate and malicious traffic since it depends on statistical isolation of anomalies rather than learning explicit class boundaries. This limitation is especially noticeable in datasets where anomalies are not clearly distinguished from typical trends.

These findings support the idea that supervised models continue to be very effective in the presence of high-quality labelled data. However, their application in dynamic contexts where new or developing threats are constantly appearing is limited by their reliance on labelled attack samples.

5.3 ANALYSIS OF ODD MODELS IN A TWO-CLASS ENVIRONMENT:

The analysis of deep learning models in a Two-class environment sheds light on how representation learning might improve intrusion detection capabilities. Deep learning techniques, in contrast to conventional tabular models, work with altered feature representations, making it possible for them to more successfully identify underlying structures in the data.

Because they can learn compact latent representations of network traffic, reconstruction-based models, Autoencoder and Variational Autoencoder in particular, perform well. These models gain from their reconstruction target, which promotes learning significant feature patterns instead of depending only on class separation, even in a Two-class environment.

DeepSVDD's relatively poorer performance suggests the view that hypersphere-based methods are less successful in Two-class settings. DeepSVDD's goal of minimising the distance to a central representation might not fully take advantage of the availability of labelled attack data because it is mainly designed for one-class learning. This restricts its capacity to attain the same degree of discrimination as models based on reconstruction. Overall, the results show that deep learning models may efficiently learn from tabular data represented by images, especially when complicated feature interactions are captured by reconstruction processes.

5.4 ANALYSIS OF ODD MODELS IN THE ONE-CLASS ENVIRONMENT:

The one-class experimental setting provides a more realistic evaluation of intrusion detection systems, as it reflects scenarios where labelled attack data is unavailable during training. In this context, the ability of models to learn normal behaviour and detect deviations becomes critical.

In this context, reconstruction-based models show definite benefits. Autoencoder and Variational Autoencoder methods create an implicit border of normalcy by learning to reconstruct normal network traffic. Effective anomaly detection is made possible by the greater reconstruction error that follows any deviation from this learned pattern.

Variational Autoencoder's excellent performance emphasises the significance of probabilistic representation learning even more. VAE can better detect tiny anomalies by modelling a distribution in the latent space, which helps it reflect variability in typical traffic.

On the other hand, even though DeepSVDD was created especially for one-class classification, it exhibits greater performance variability. This implies that hypersphere-based techniques are extremely sensitive to feature representation and may have trouble with complicated or poorly clustered underlying data distributions. These results highlight how reconstruction-based methods offer a more adaptable and consistent framework for anomaly detection when normal data is all that is available.

5.5 CROSS-DATASET GENERALIZATION AND MODEL ROBUSTNESS:

A key aspect of evaluating intrusion detection algorithms is their ability to perform consistently across diverse datasets. In this work, the evaluated models were tested on two intrusion detection datasets: SDN20 and BAIoT Mirai, which have different traffic characteristics and attack behaviours.

The findings show that reconstruction-based deep learning models, specifically the Autoencoder and Variational Autoencoder, consistently outperformed both datasets. These models were able to effectively understand the fundamental structure of normal network data and identify anomalous or malicious activity. Their capability to reconstruct normal data patterns allowed them to generalize effectively across a variety of network settings.

In contrast, the hypersphere-based model DeepSVDD performed quite poorly in several studies. While the model could learn representations of normal network traffic, its capacity to distinguish between normal and anomalous patterns was more dependent on the dataset's properties. This resulted in less consistent performance than reconstruction-based techniques.

When the overall behaviour of the examined models was compared across both datasets, the reconstruction-based techniques proved to be more resilient and stable. These models retained good detection performance regardless of the dataset utilised, showing that they are well-suited for intrusion detection scenarios in which network traffic patterns fluctuate dramatically.

Overall, the findings indicate that reconstruction-based anomaly detection methods are a more accurate and consistent framework for identifying network intrusions when applied to DeepInsight-generated visual representations of tabular data. Their capacity to generalise across datasets demonstrates their promise for practical use in real-world intrusion detection systems.

5.6 OVERALL RESEARCH INSIGHTS:

Based on the complete experimental evaluation, several major study insights emerge: First, when high-quality labelled data is available, traditional supervised models retain their extraordinary strength.

Second, while the DeepInsight transformation is excellent at enabling deep learning on structured network traffic, it does not consistently outperform strong tabular baselines.

Third, reconstruction-based deep models, particularly the Variational Autoencoder, deliver the most reliable and consistent results in one-class anomaly detection circumstances.

Fourth, DeepSVDD, while conceptually appealing, is moderately sensitive to representation quality and may require additional architectural modification for optimal performance.

Finally, the findings indicate that hybrid intrusion detection systems combining supervised classification and deep OOD detection may be the most practicable approach for real-world cybersecurity deployments.

5.7 CONCLUSION

Intelligent and adaptive intrusion detection systems have become increasingly important due to the rapid development of cloud computing, networked systems, and Internet of Things (IoT) environments. Modern cyberthreats, especially zero-day attacks and previously unseen incursion patterns, cannot be handled by traditional signature-based methods. In this regard, deep learning and machine learning approaches have shown promise as ways to improve intrusion detection capabilities.

Using DeepInsight image transformation, this thesis examined the efficacy of Out-of-Distribution (OOD) algorithms for network intrusion detection, with an emphasis on contrasting deep learning techniques with traditional tabular models. The goal of the study was to determine whether, in practical intrusion detection scenarios, modern deep OOD algorithms offer quantifiable benefits over conventional tabular methods.

A thorough experimental framework was created and tested on two typical datasets, SDN20 and BAIoT_mirai, in order to accomplish this goal. Both supervised and One-Class learning paradigms were used in the methodology, which comprised deep OOD models (Autoencoder, Variational Autoencoder, and DeepSVDD) and classical machine learning models (Random Forest and Isolation Forest).

The results of the experiment showed a number of significant conclusions.

First, on both datasets, Random Forest continuously performed almost flawlessly in the Two-Class setting. This demonstrates that when labelled attack samples are available, ensemble tree approaches continue to be very powerful baselines for structured intrusion detection data. Strong model stability under the assessed conditions is also indicated by the small generalization gap between training and testing.

Second, as compared to supervised techniques, Isolation Forest showed reasonable but less performance. Although it works well as an unsupervised baseline model, models that learn richer feature representations have a greater detection power. This emphasizes how crucial deep representation learning is for sophisticated network behavior.

Third, the application of convolutional architectures to tabular network traffic data was successfully made possible by the DeepInsight transformation. When trained on DeepInsight images, Autoencoder and Variational Autoencoder models performed exceptionally well, frequently similar to a supervised tabular classifier. This demonstrates that significant relationships within structured network data can be preserved by spatial feature mapping.

Fourth, the Autoencoder and Variational Autoencoder showed strong and consistent performance in the One-Class environment, which more closely represents actual intrusion detection situations. These models are good choices for situations when labelled attack data is hard to obtain or unavailable because they successfully detected abnormal traffic using reconstruction-based criteria.

Fifth, the behavior of DeepSVDD was inconsistent. Its performance declined in the One-Class experiments, despite being competitive in the supervised scenario. The findings imply that feature distribution, threshold selection, and representation quality have an impact on hypersphere-based techniques. This observation is consistent with prior literature and highlights the need for careful tuning when deploying DeepSVDD in practical environments. This finding is in line with other research and emphasizes the necessity of precise adjustment when implementing DeepSVDD in real-world scenarios.

Overall, the results show that while image-based OOD techniques show promise, they are not always superior to traditional tabular techniques. Yet, deep anomaly detection models, in particular, Autoencoder and Variational Autoencoder, offer useful complementary features in practical situations where new attacks are constantly emerging and labelled data is limited.

5.8 KEY CONTRIBUTION:

The following concisely describes the primary contributions of this thesis:

1. **Comprehensive Comparative Analysis:**

   For intrusion detection, a thorough comparison of deep OOD techniques and traditional tabular models was conducted.

2. **IDS Pipeline Based on DeepInsight:**

   To convert tabular network traffic into visual representations appropriate for CNN-based models, an end-to-end pipeline was implemented.

3. **Evaluation of Several OOD Models:**

   Both One-Class and Two-Class paradigms were used to execute and evaluate Autoencoder, Variational Autoencoder, and DeepSVDD.

4. **Validation across datasets:**

   Two different datasets (SDN20 and BAIoT_mirai) were used in the experiments, thereby improving the generalization and robustness of the findings.

5.9 USEFUL RECOMMENDATIONS:

The following suggestions for practical intrusion detection deployments can be made in the context of the experimental results:

- Classical supervised models, like Random Forest, continue to be successful and computationally efficient when high-quality labelled attack data are available.

- Reconstruction-based deep models (AE, VAE) offer a more robust anomaly detection capability in environments where new or unidentified attacks are expected to occur.

- Although DeepInsight transformation is a practical method for applying deep learning to tabular intrusion data, robust tabular classifiers shouldn't be automatically replaced.

- For optimal practical performance, hybrid systems that combine supervised and OOD detection may be the best option.

5.10 LIMITATIONS:

Although this study offers insightful information, there are still a number of Limitations.

- One limitation is still dependent on the original network features' quality. All models' performance is still dependent on the input feature set's discriminative power, even though DeepInsight automates spatial transformation.

- Dimensionality reduction through UMAP is introduced by the DeepInsight transformation, which could result in incomplete persistence of high-dimensional feature associations and partial information loss.

- When there are a lot of tabular features, DeepInsight's fixed image resolution could result in feature overlap or spatial compression.

- Moreover, this work's deep structures were comparatively light. Performance on DeepInsight pictures may be further enhanced with more sophisticated backbones like ResNet or Vision Transformers.

- Since deep models, especially DeepSVDD and VAE, are sensitive to hyperparameter selection (such as latent dimension, learning rate, and threshold percentile), thorough tuning was beyond the scope of this research.

- Scalability may be limited in high-speed or resource-constrained network contexts due to the computational expense added by DeepInsight and convolutional models.

- Since the datasets are flow-based, they might not accurately depict environments where encrypted traffic dominates, which restricts direct generalization to scenarios involving completely encrypted networks.

5.11 FUTURE WORK:

This work offers several promising paths for future research.

- An important avenue to improve One-Class stability is to investigate soft-boundary DeepSVDD and related versions that automatically learn the hypersphere radius.

- Developing hybrid tabular-image fusion models, which combine the positive aspects of deep spatial representation and traditional feature-based learning, is an additional approach.
- Future research can potentially look into transformer-based intrusion detection architectures, which have recently demonstrated excellent performance in the tabular and vision domains.
- Furthermore, adding concept drift adaptation and continuous learning would strengthen robustness in dynamic network settings where traffic distributions change over time.
- Streaming evaluation and real-time deployment are two further crucial steps in developing OOD-based IDS solutions for production.
- Finally, the interpretability and reliability of deep intrusion detection systems, both of which are essential for cybersecurity operations, may be enhanced by using explainable AI methods.

5.12 FINAL REMARKS:

This thesis concludes by showing that deep Out-of-Distribution techniques offer significant extra features for identifying unknown and dynamic threats, even though classical supervised models like Random Forest continue to be incredibly effective for intrusion detection when labelled data is available. Particularly, autoencoder and variational autoencoder models exhibit great potential for both real-world Two-Class and One-Class scenarios.

Future intrusion detection systems will probably profit from hybrid approaches that incorporate the advantages of deep representation learning, OOD detection techniques, and traditional machine learning as cyber threats continue to change. The results of this thesis help to clarify these trade-offs and lay the groundwork for future studies in resilient and flexible network security.

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